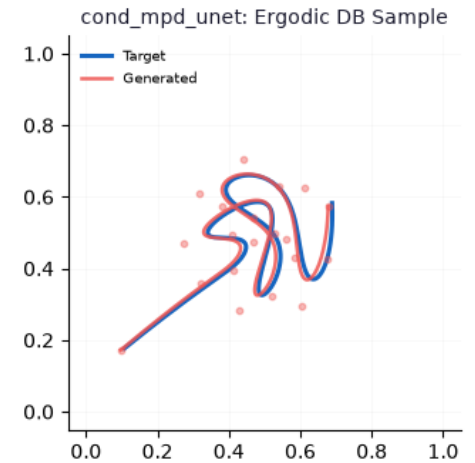
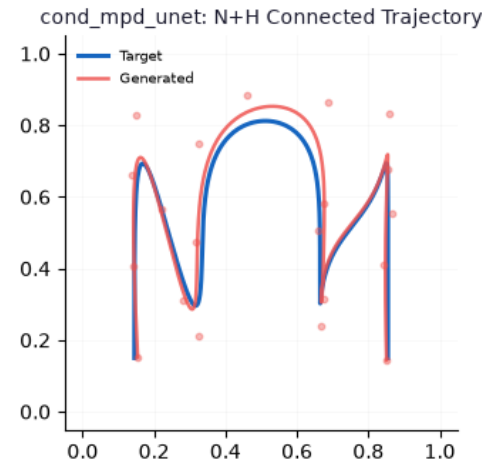
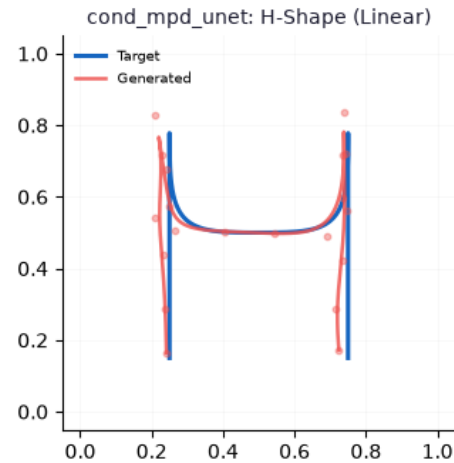
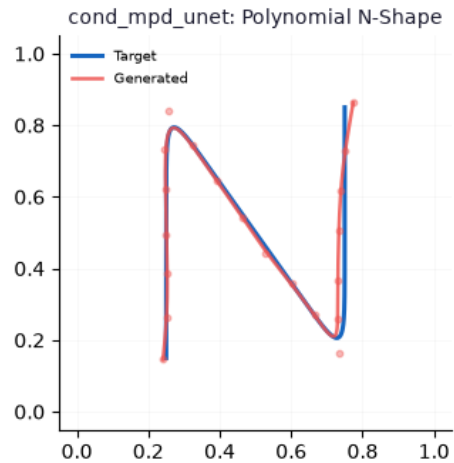
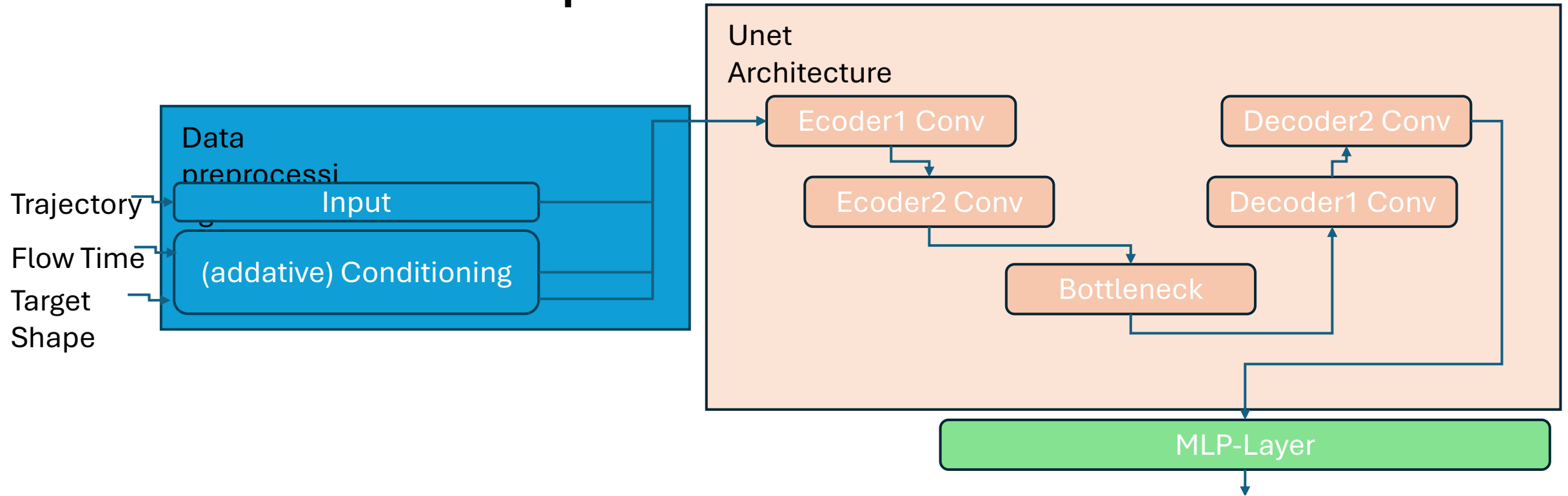


Meeting notes 10.09

Architecture developement – first version

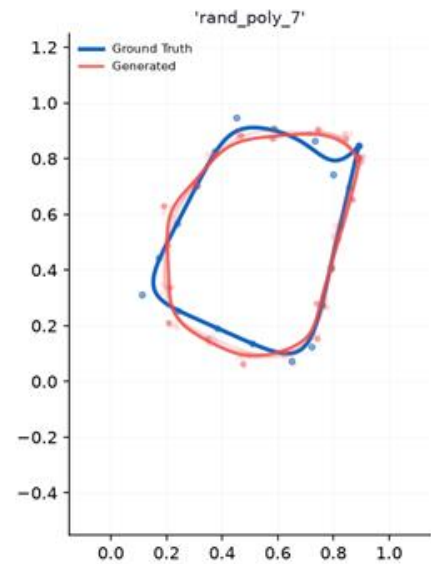
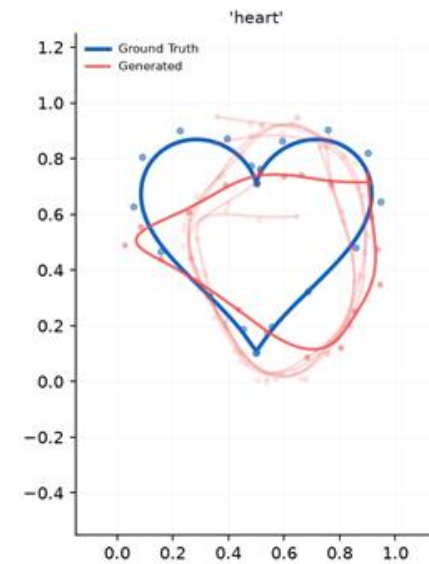
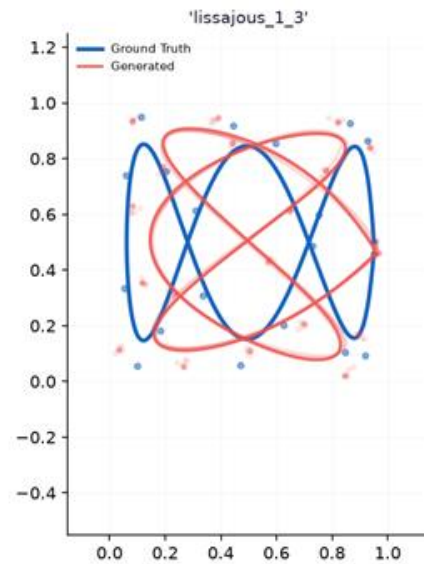
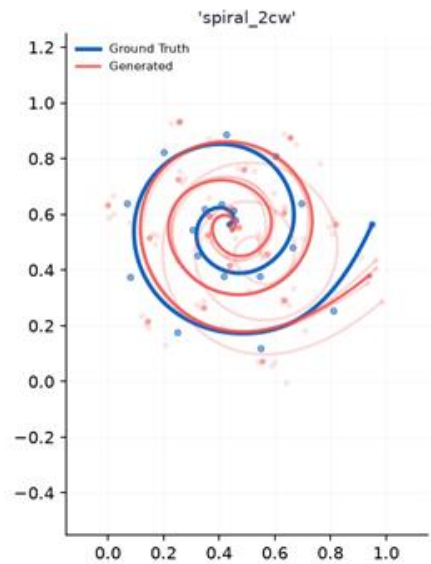
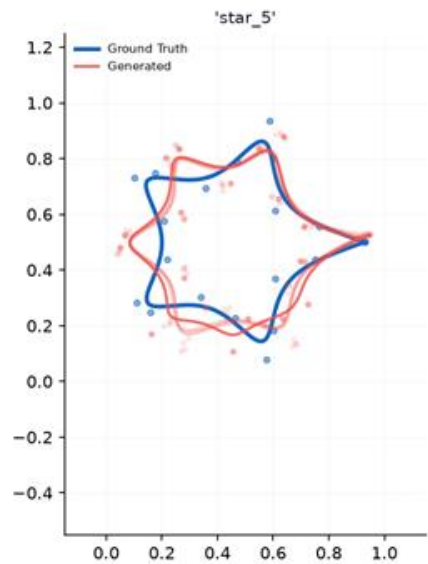
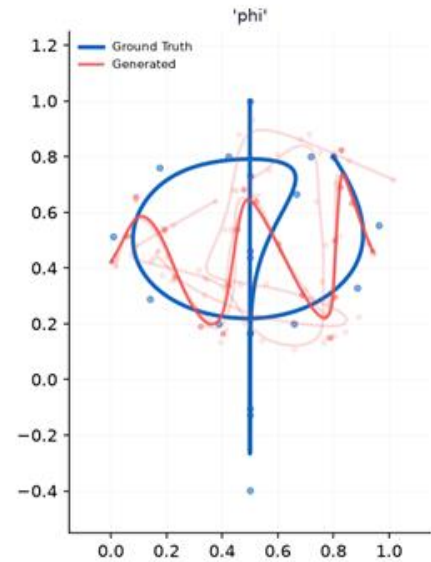
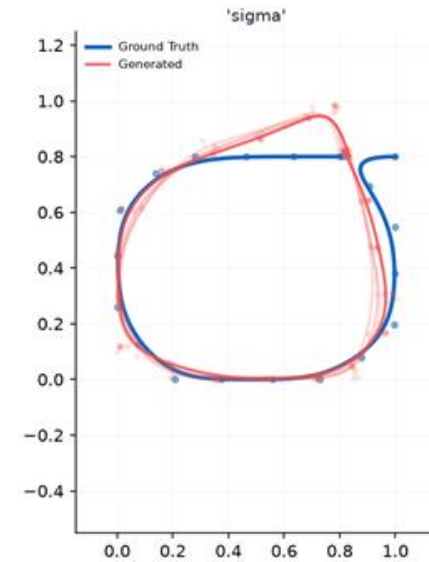
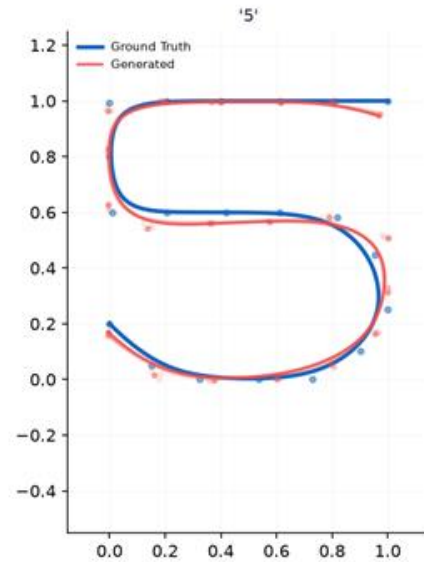
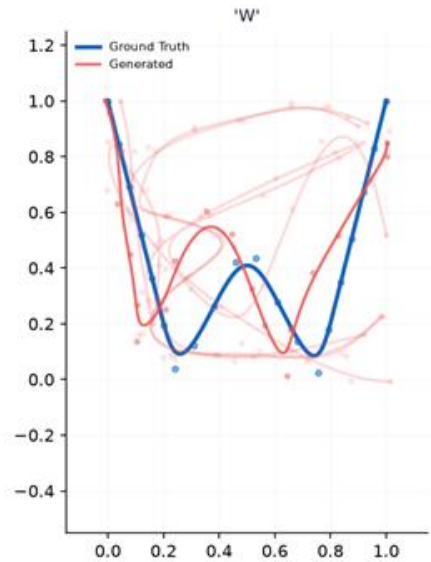
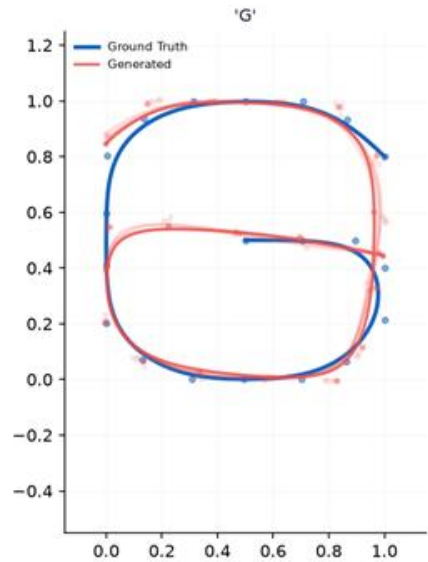


Conditioning representations

- Spectral- vs Waypointconditioning

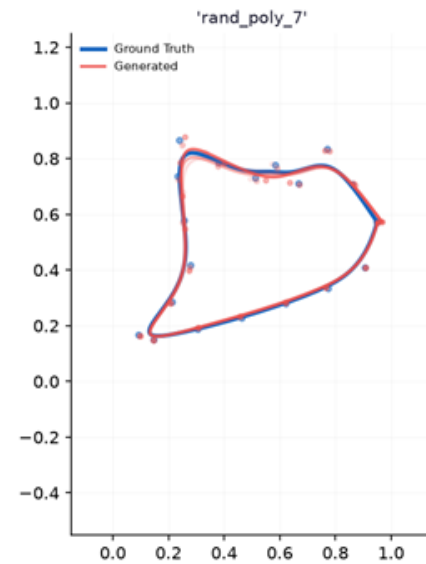
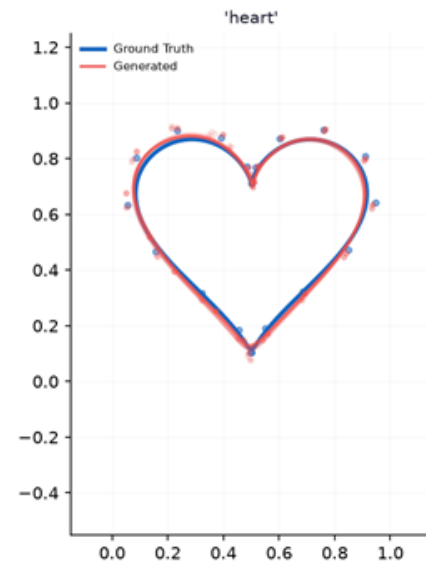
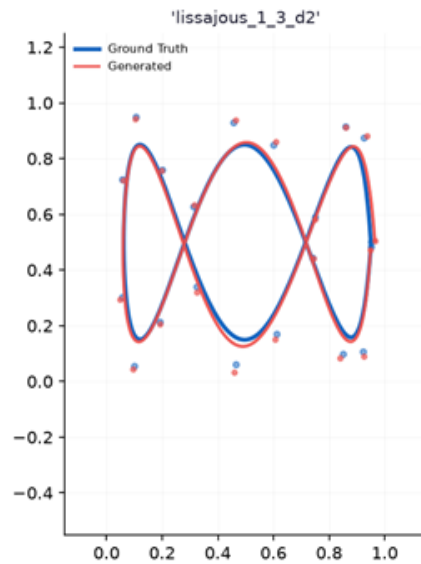
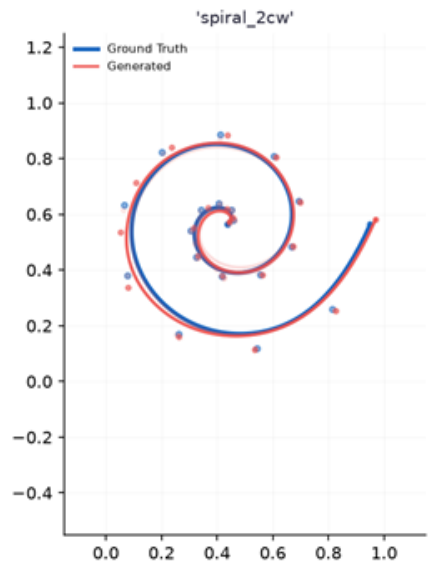
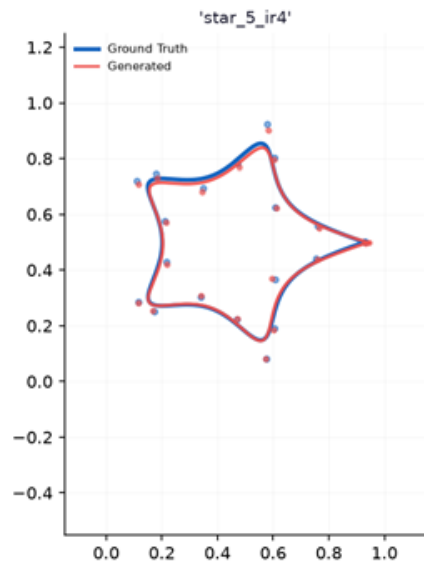
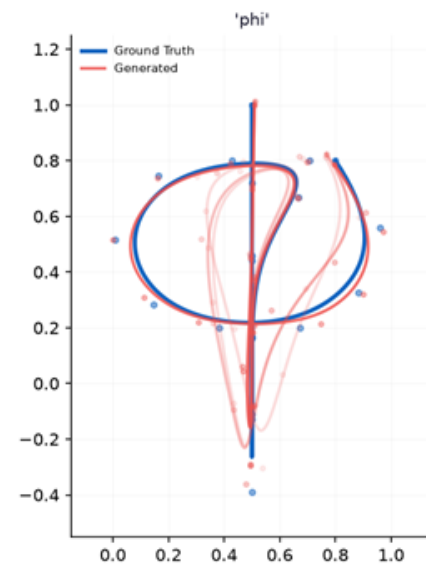
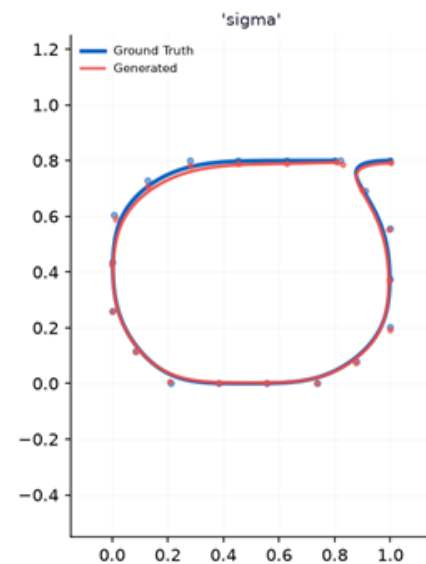
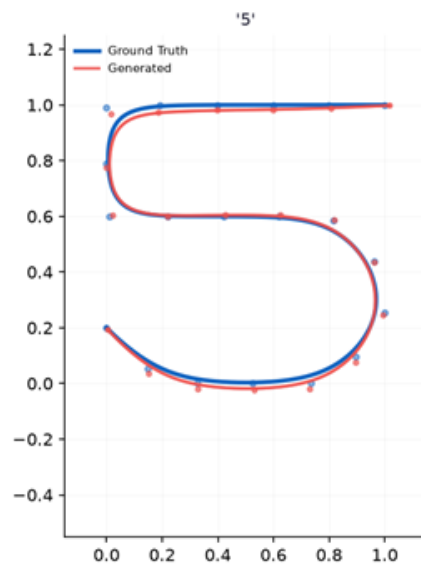
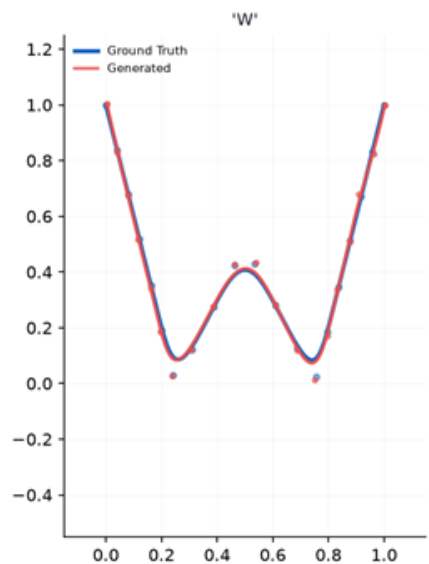
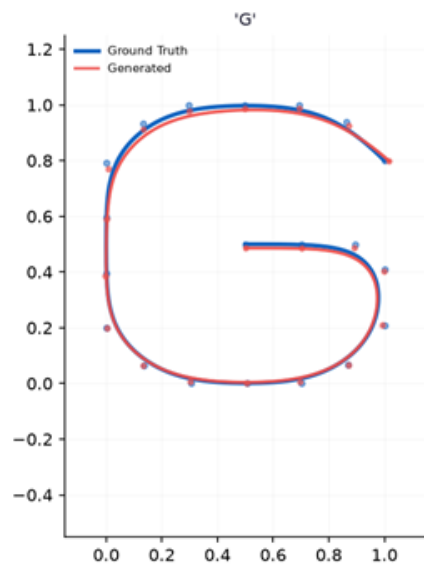
Spectral – conditioning - fourrier projection

Holdout Shapes - Epoch 1000



Waypoints conditioning

Holdout Shapes - Epoch 250



Dataset

Target Distributions & Init Trajectories (Batch 1)



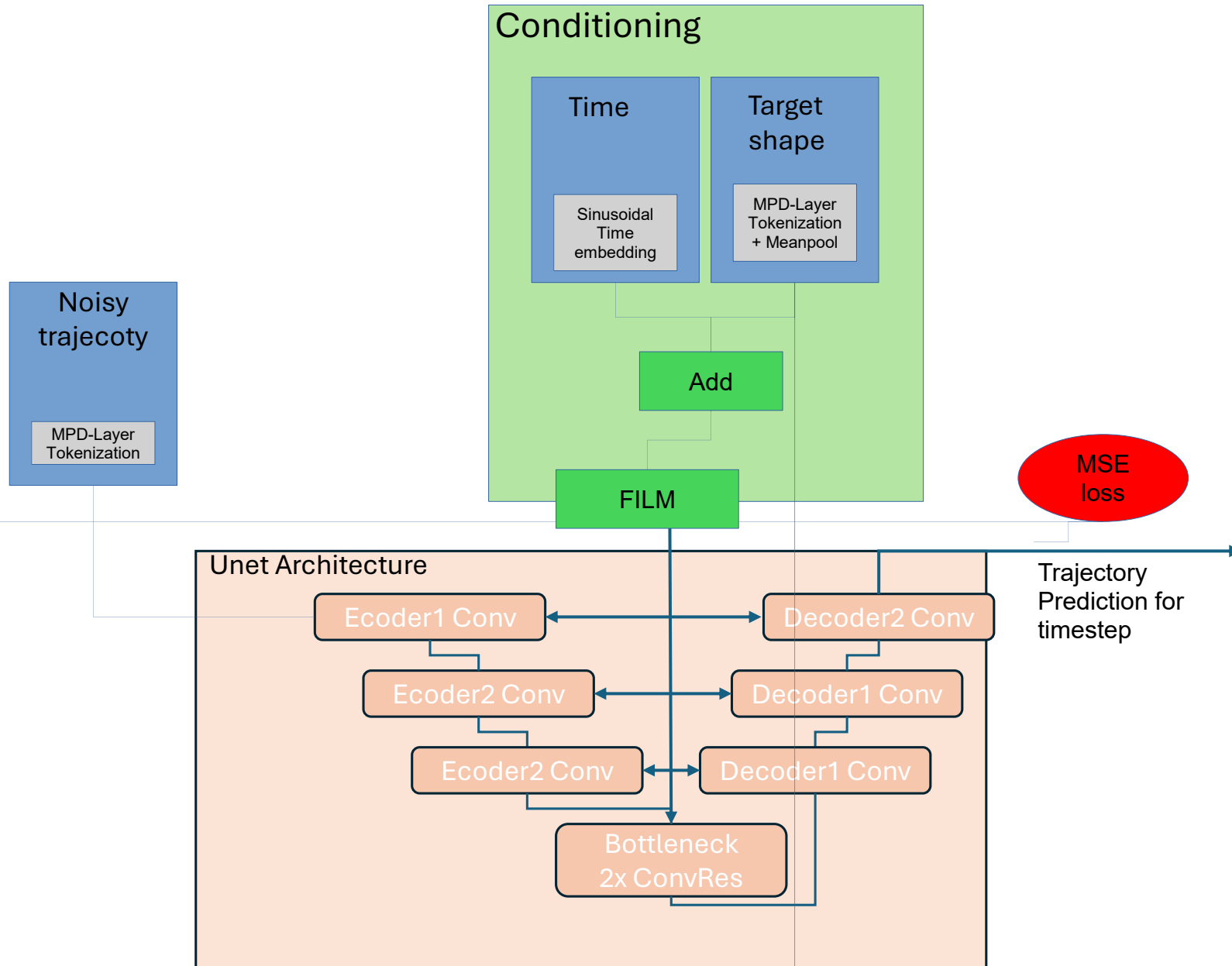
Architecture development

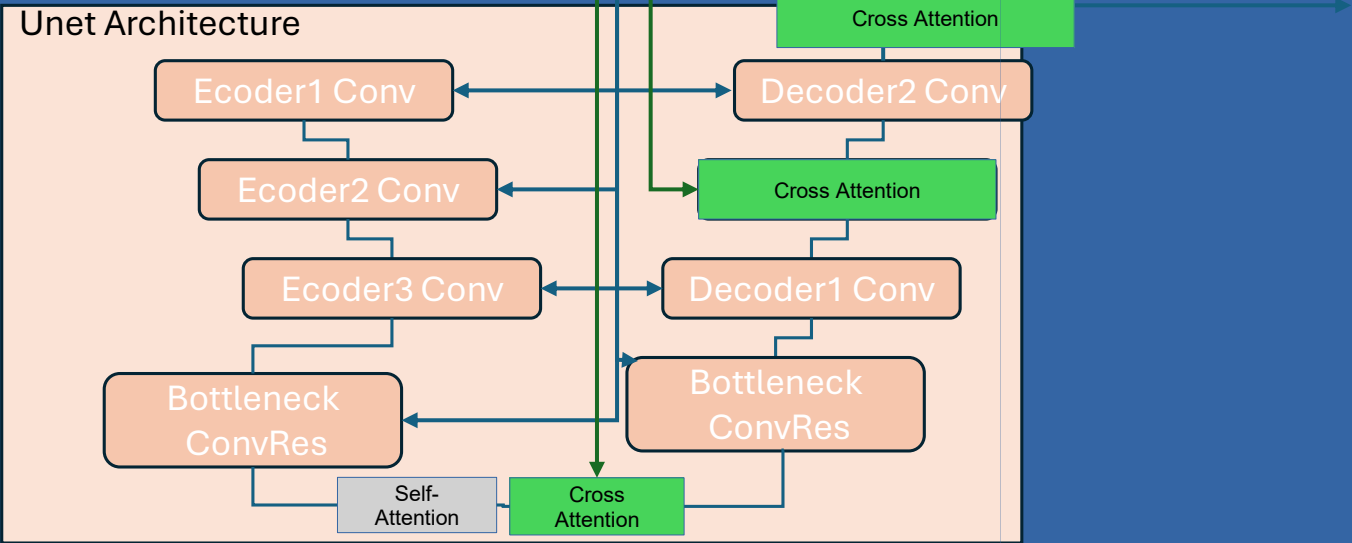
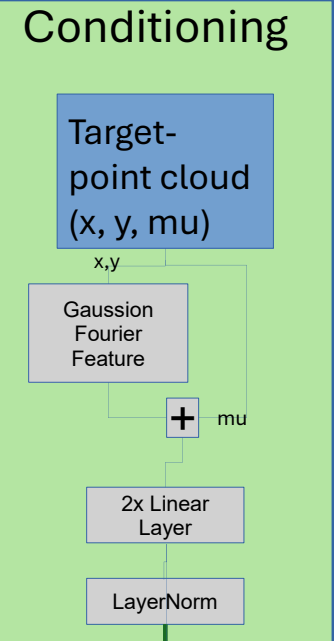
Conditioning tests

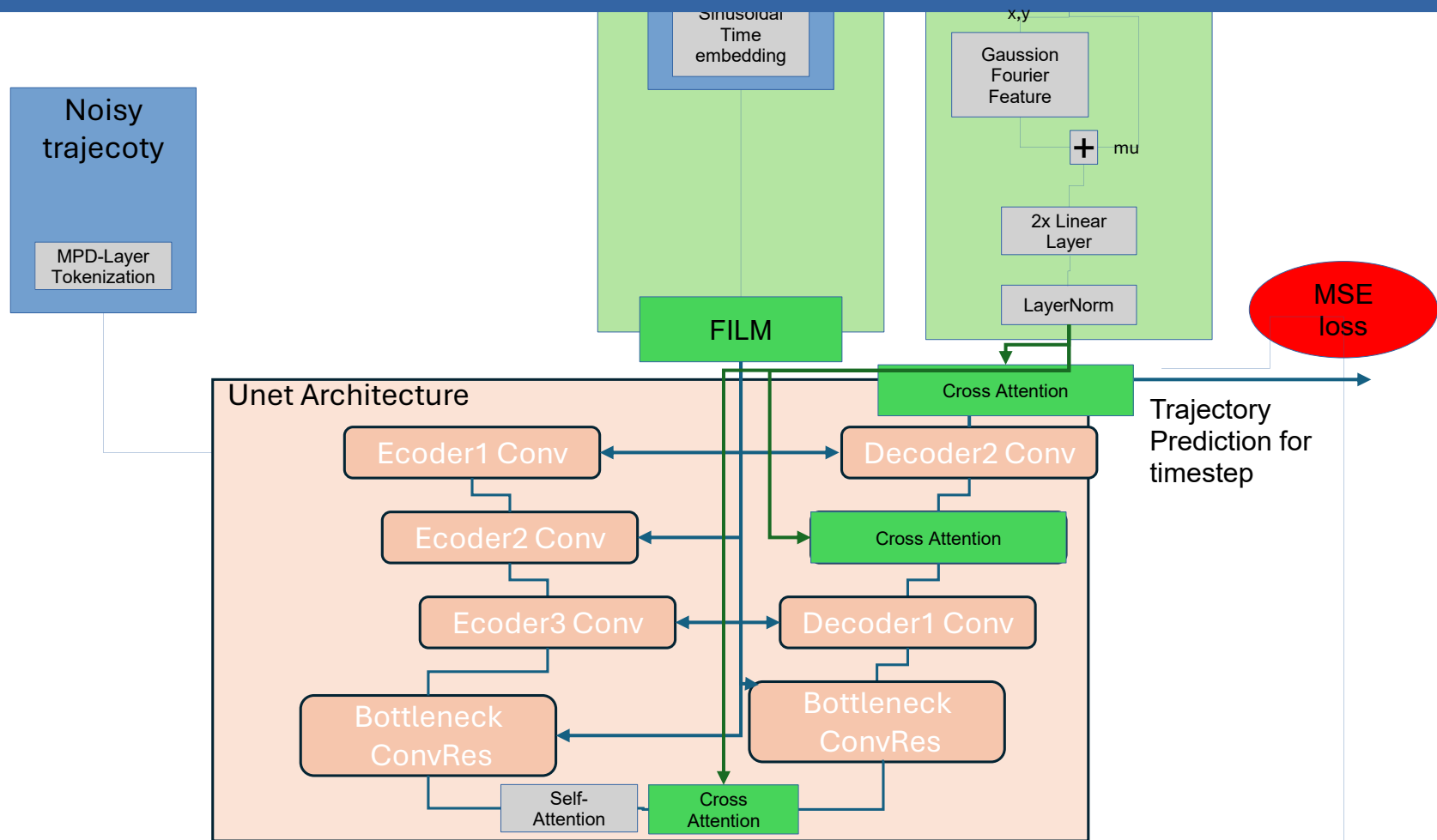
- Global average pooling + additional conditioning
- FiLM conditioning
- Cross attention based conditioning

Additive
conditioning

FILM Conditioning

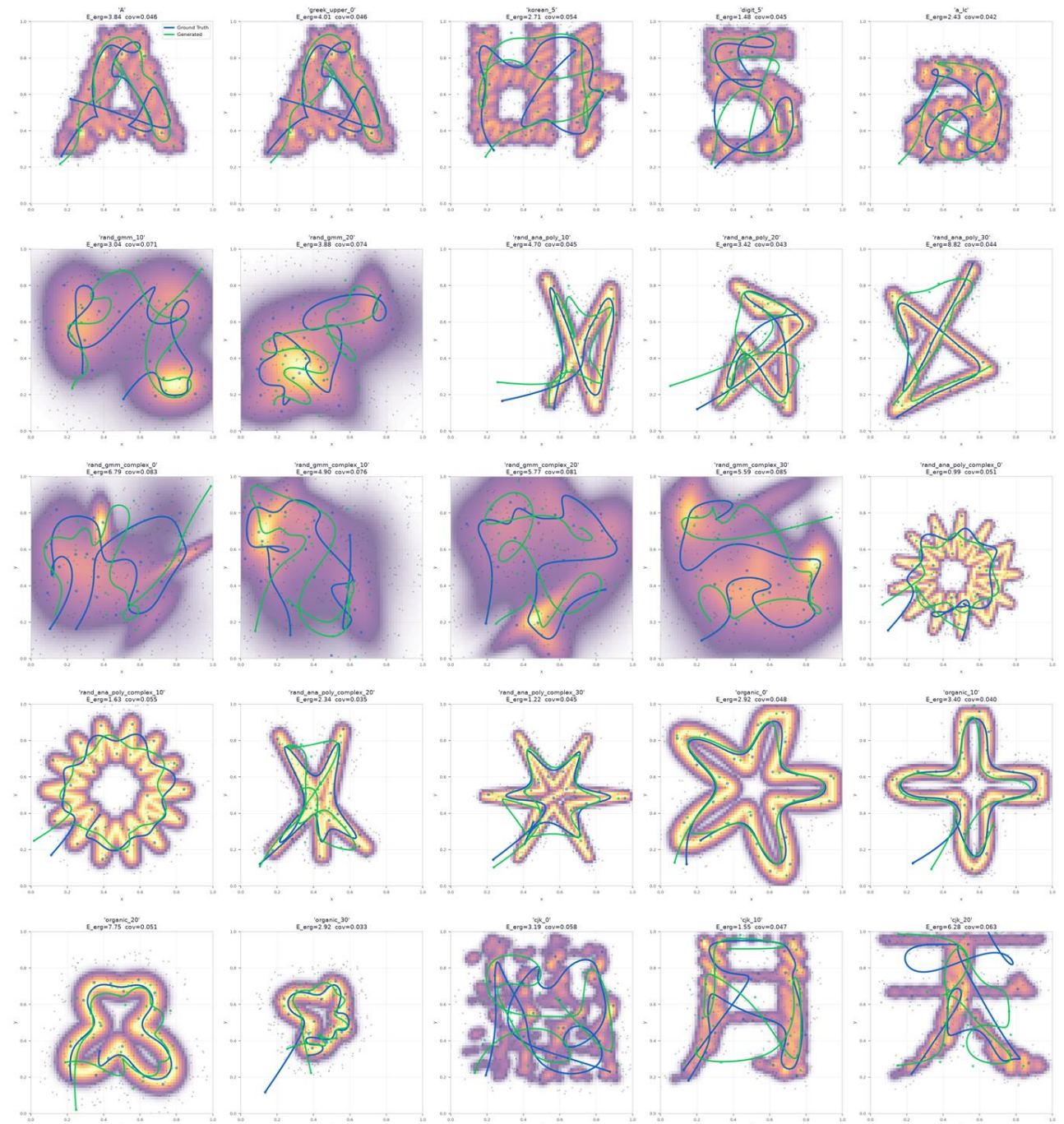




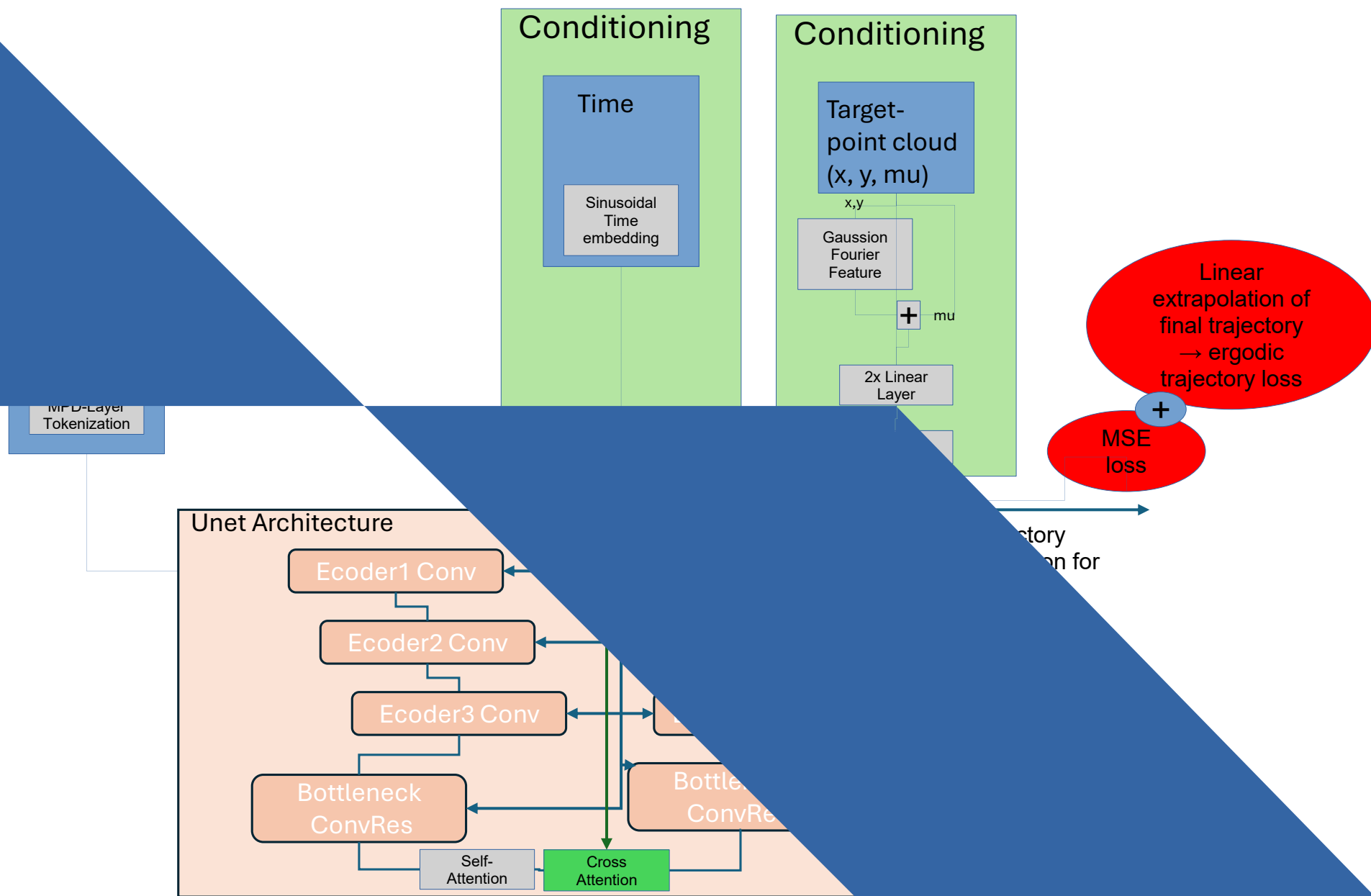


Results: flow matching

Shortest lengths costs but whorst ergodicity

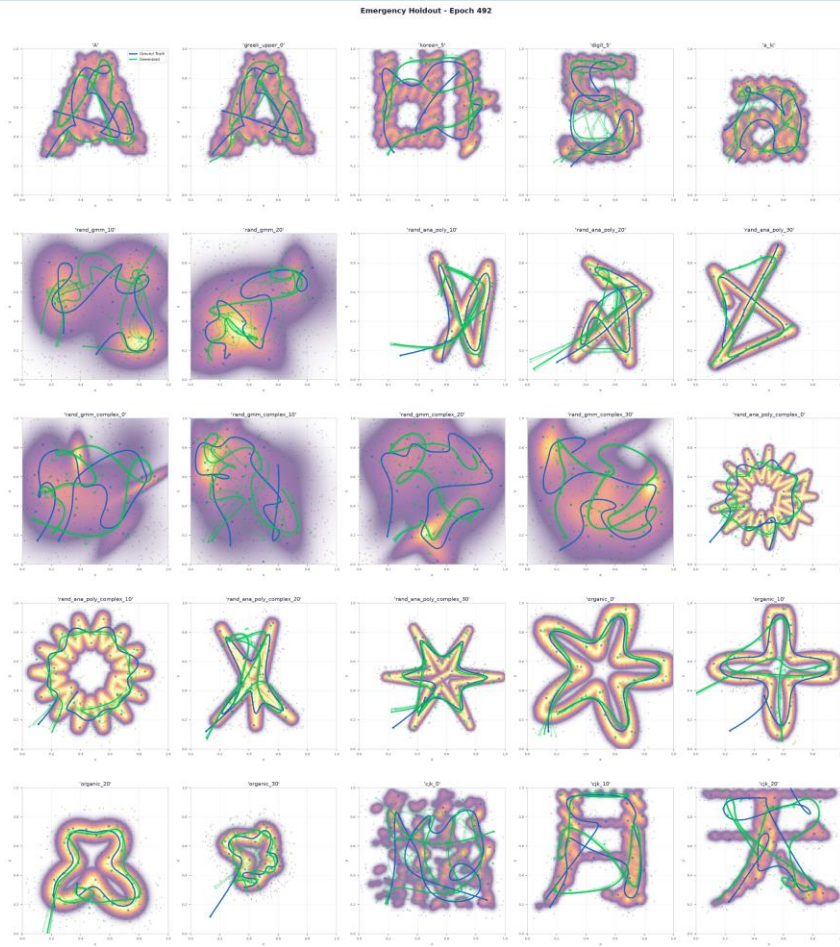


Added ergodic loss

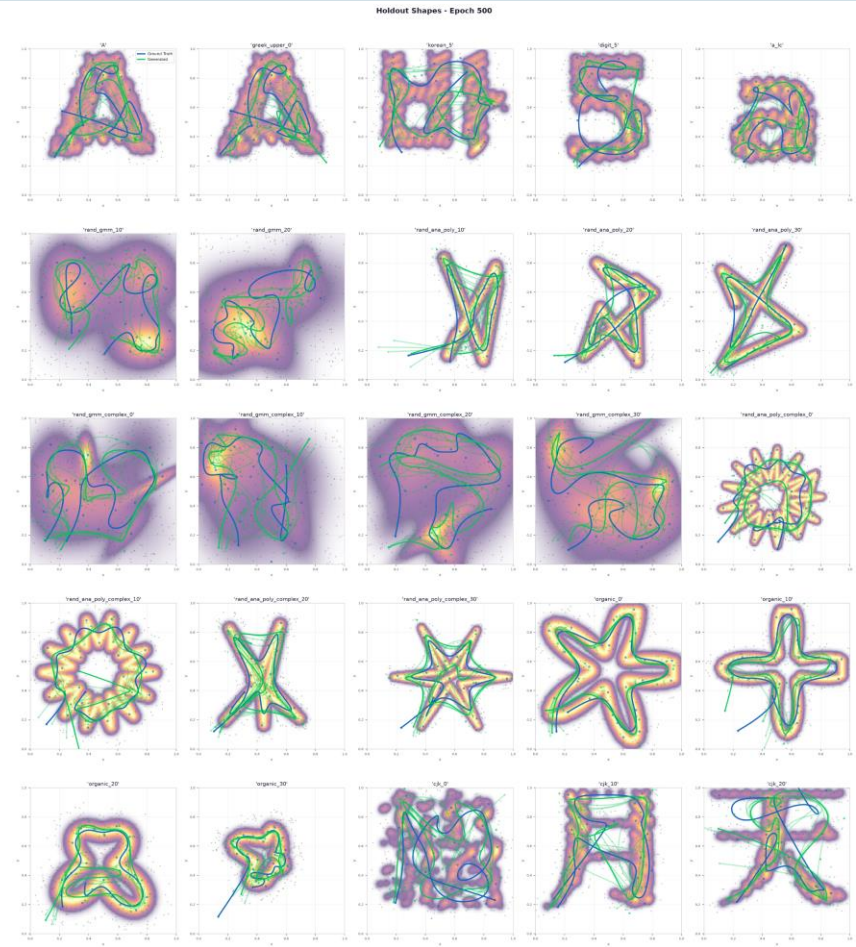


Ergodic loss terms

$\lambda = 2$



$\lambda = 400$

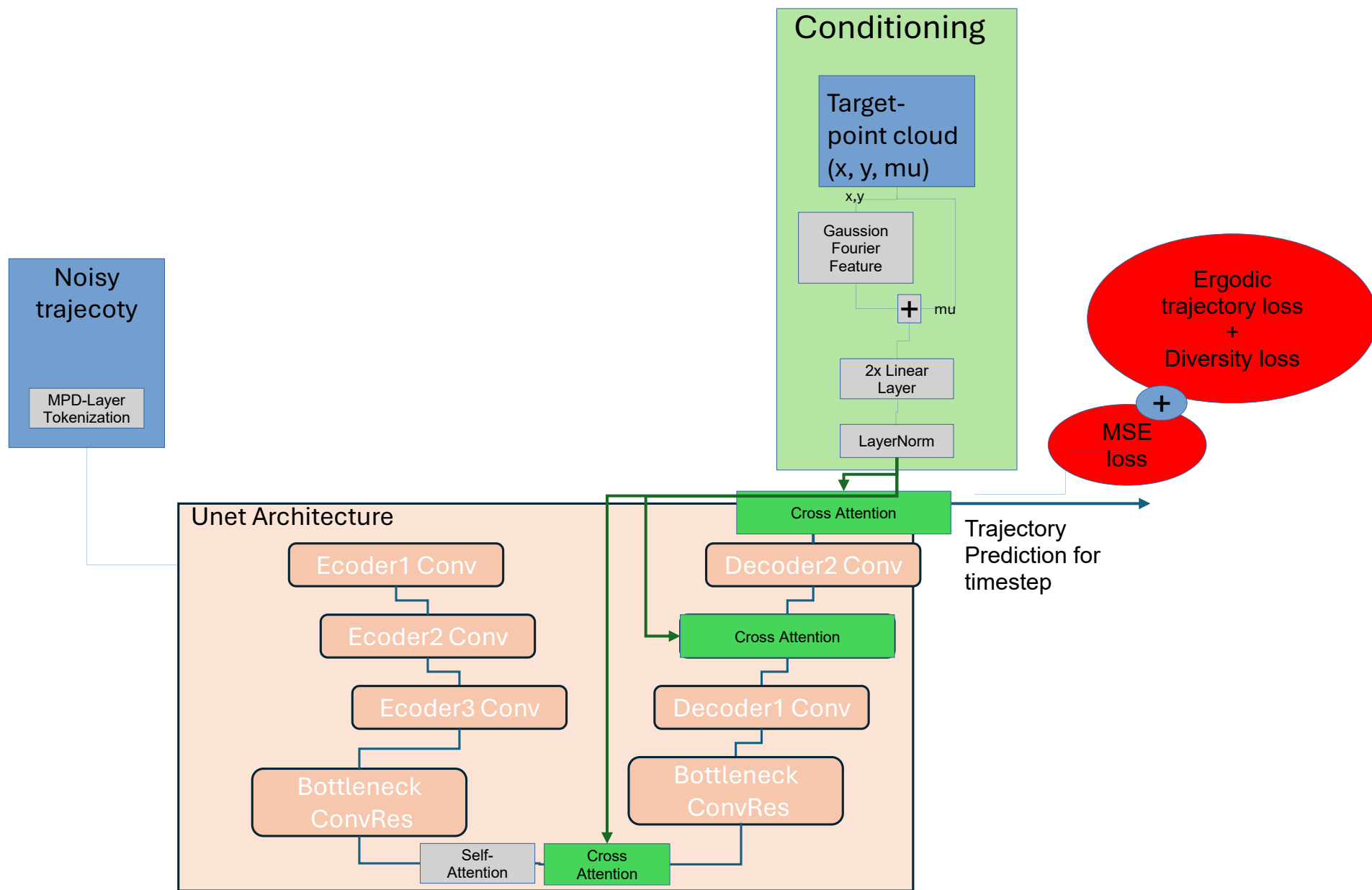


Ergodic loss terms

	Ergodic error (Fourrier)
CFM	4,651
$\lambda = 2$	4,254
$\lambda = 25$	3,158
$\lambda = 100$	2,460
$\lambda = 200$	2,245
$\lambda = 400$	2,140
SVGD-Solver (“Groundtruth” from db)	2,709

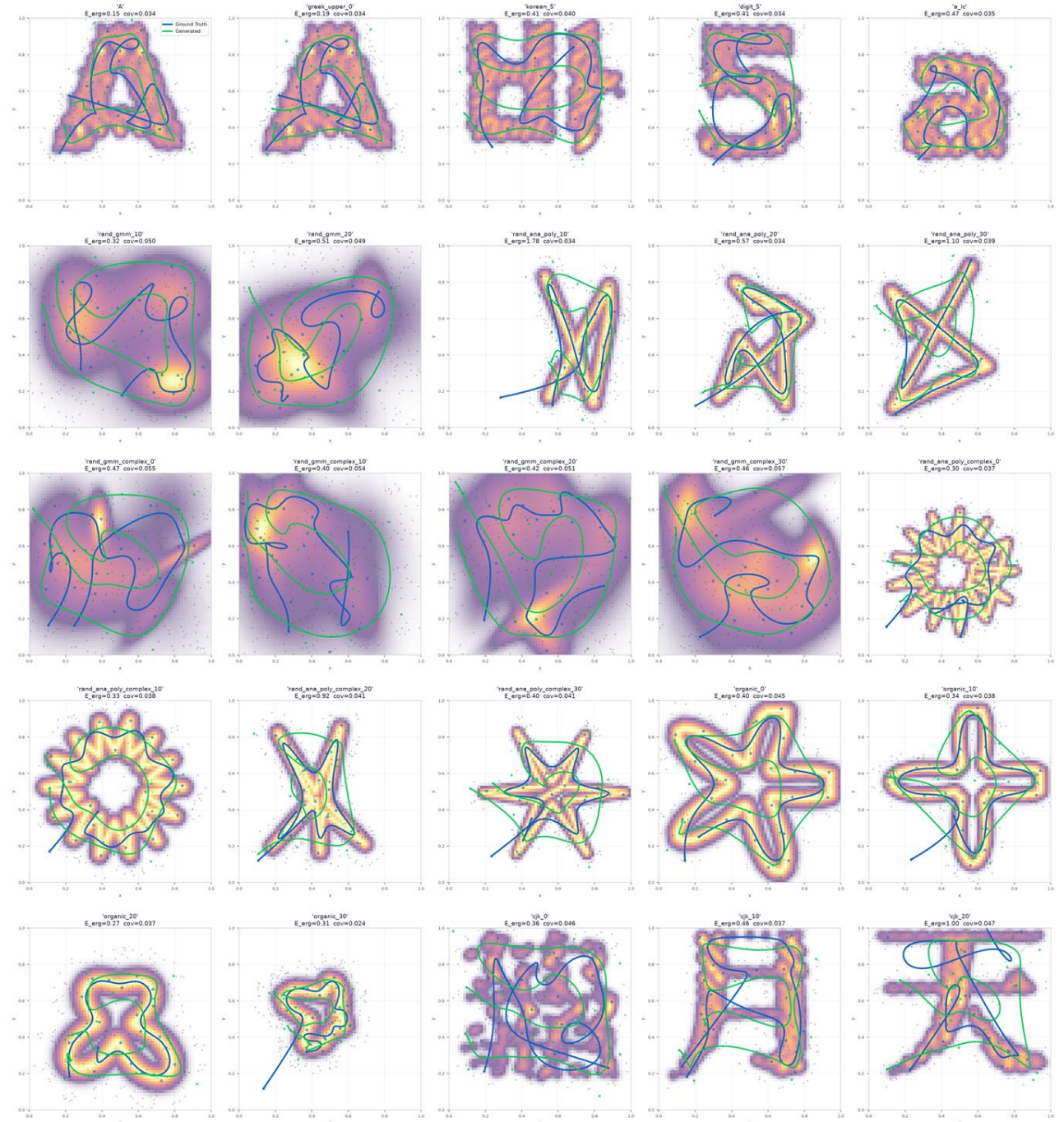
Tradeoff: Ergodic loss to length

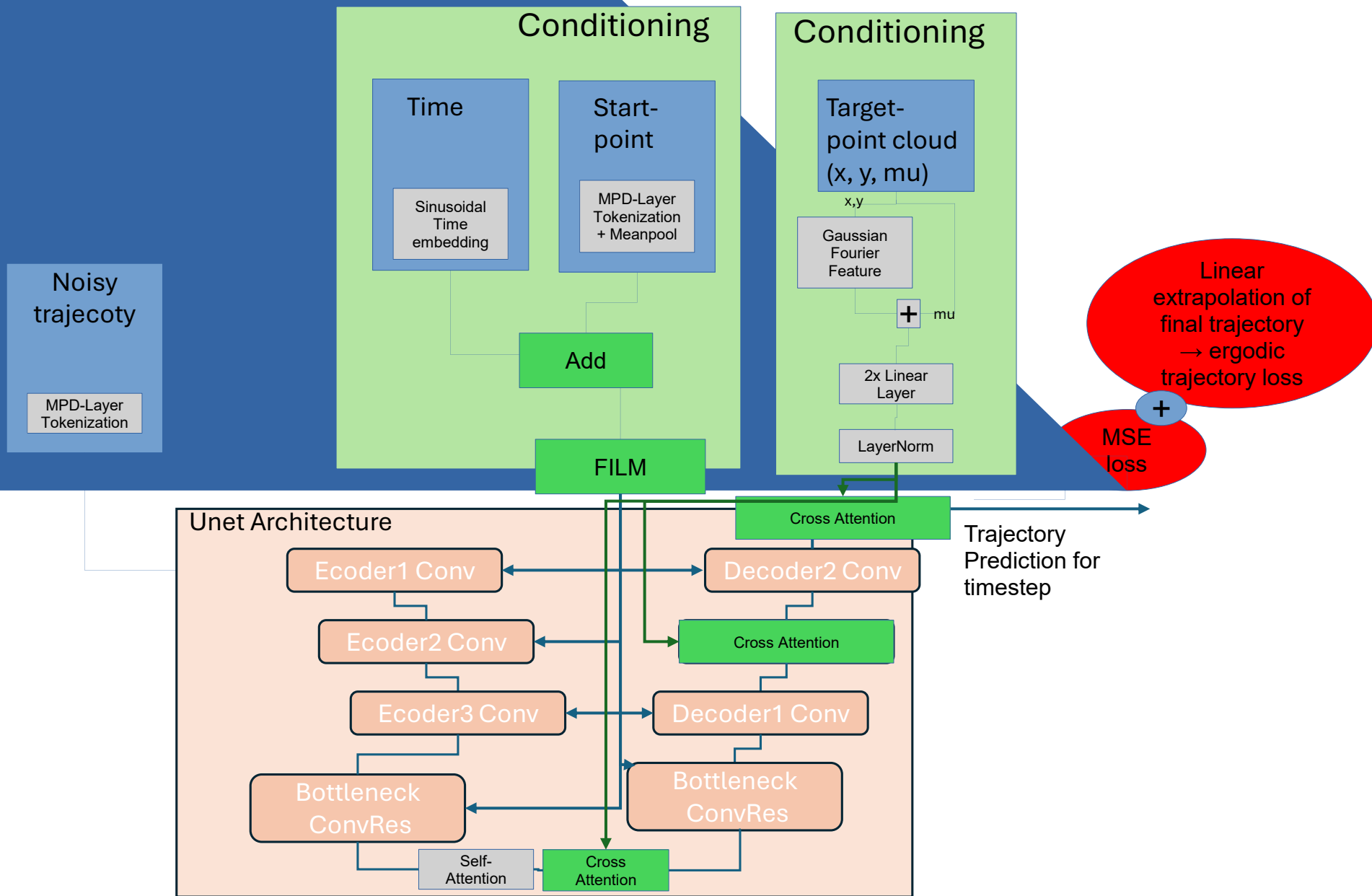
Selfsupervised only



Results: self supervised

Much higher lengths costs but better ergodicity



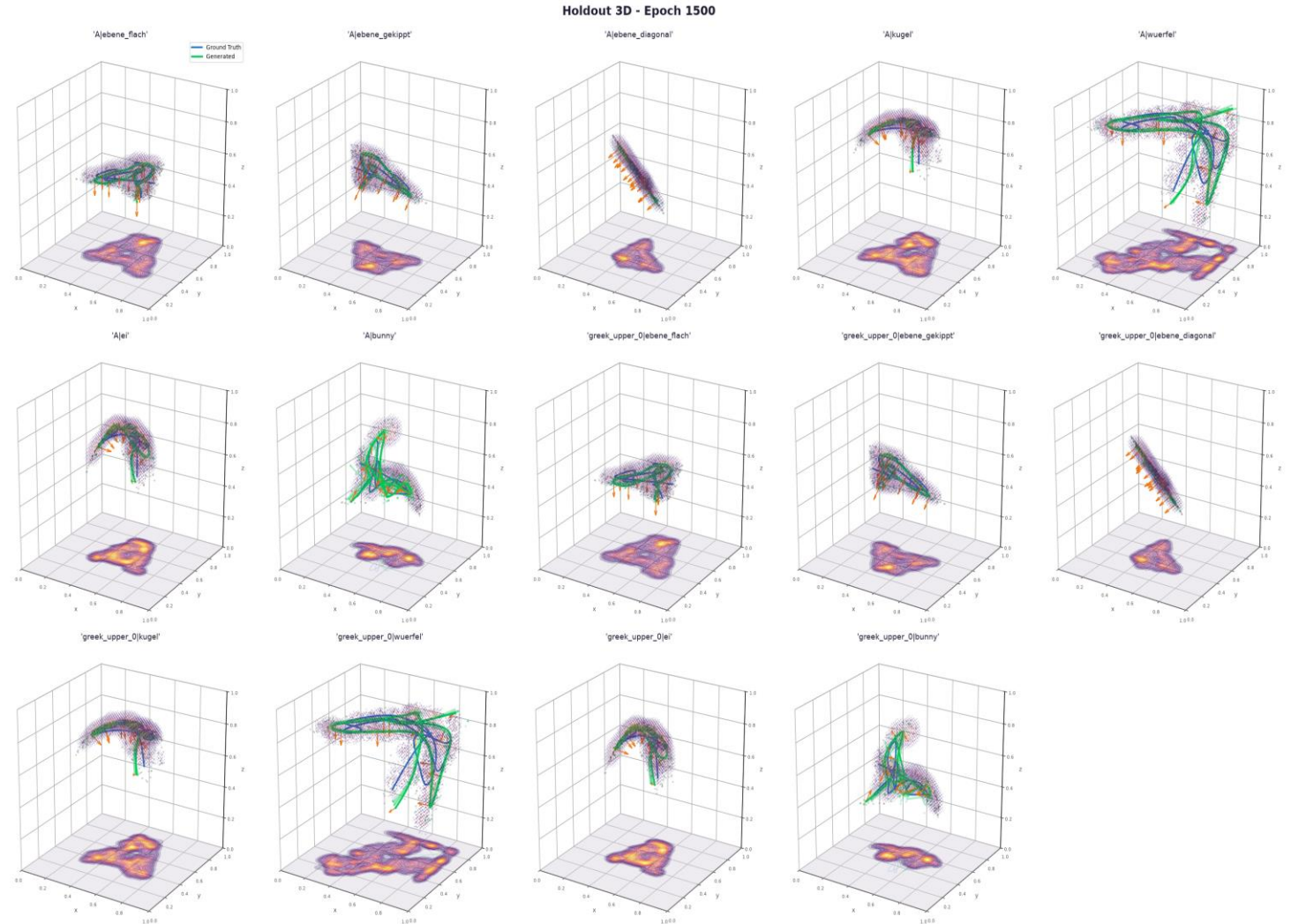


3D Transfer

- New Databank which projects the target distribution/trajectory on different shapes (natural shapes, stanford shape library)

3D transfer

- second step:
 - Change database:
 - Project shapes and trajectories on different surfaces
 - Train on new database

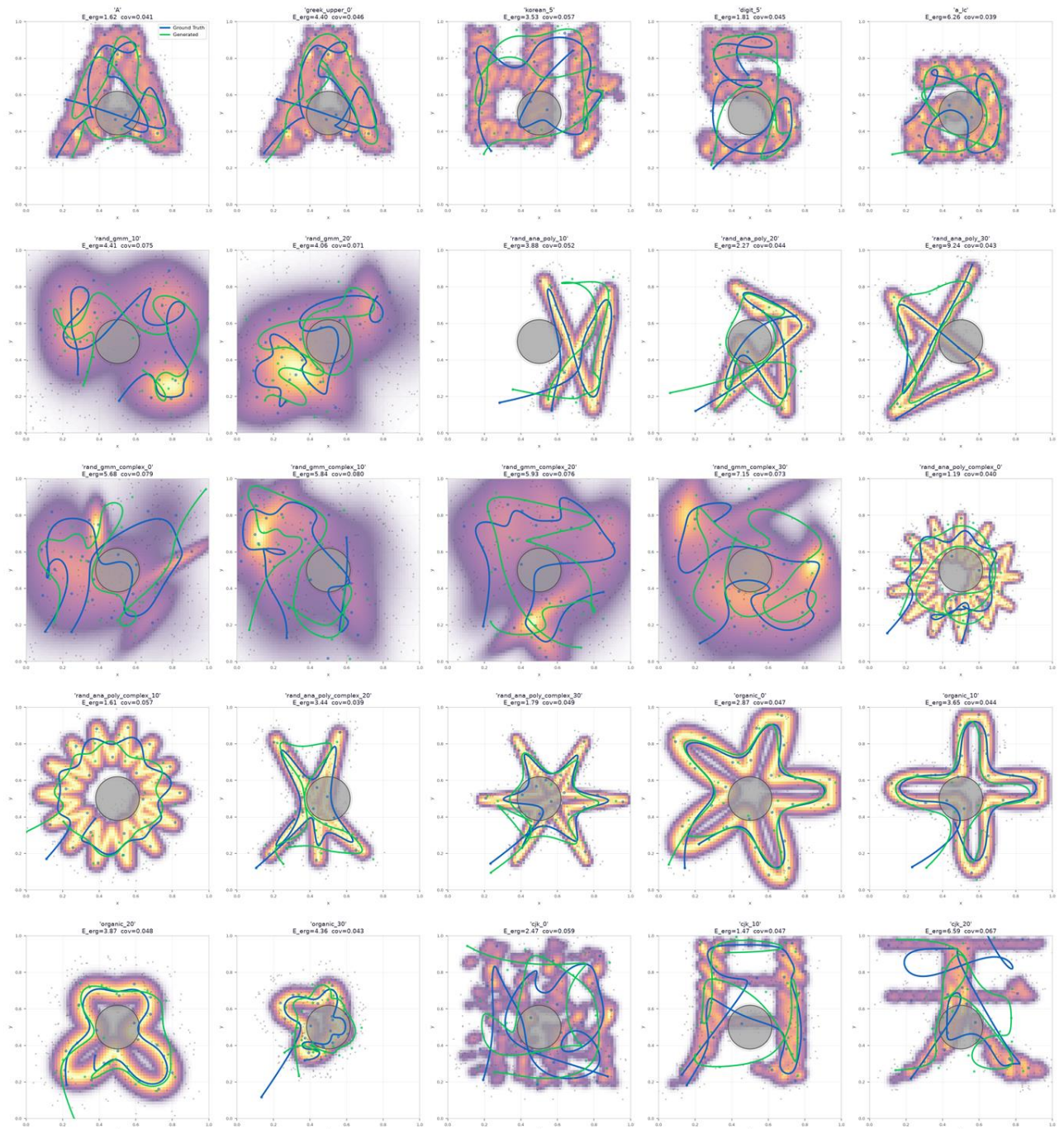


Constraint enforcement with additional forces during inference of the flow

- Obstacle constraint (already implemented)
 - Bounding box constraints
 - Fix point constraint
 - 3D object penetration constraints
 - Length constraints – still difficult
-
- General approach – enforce constraints with forces during inference

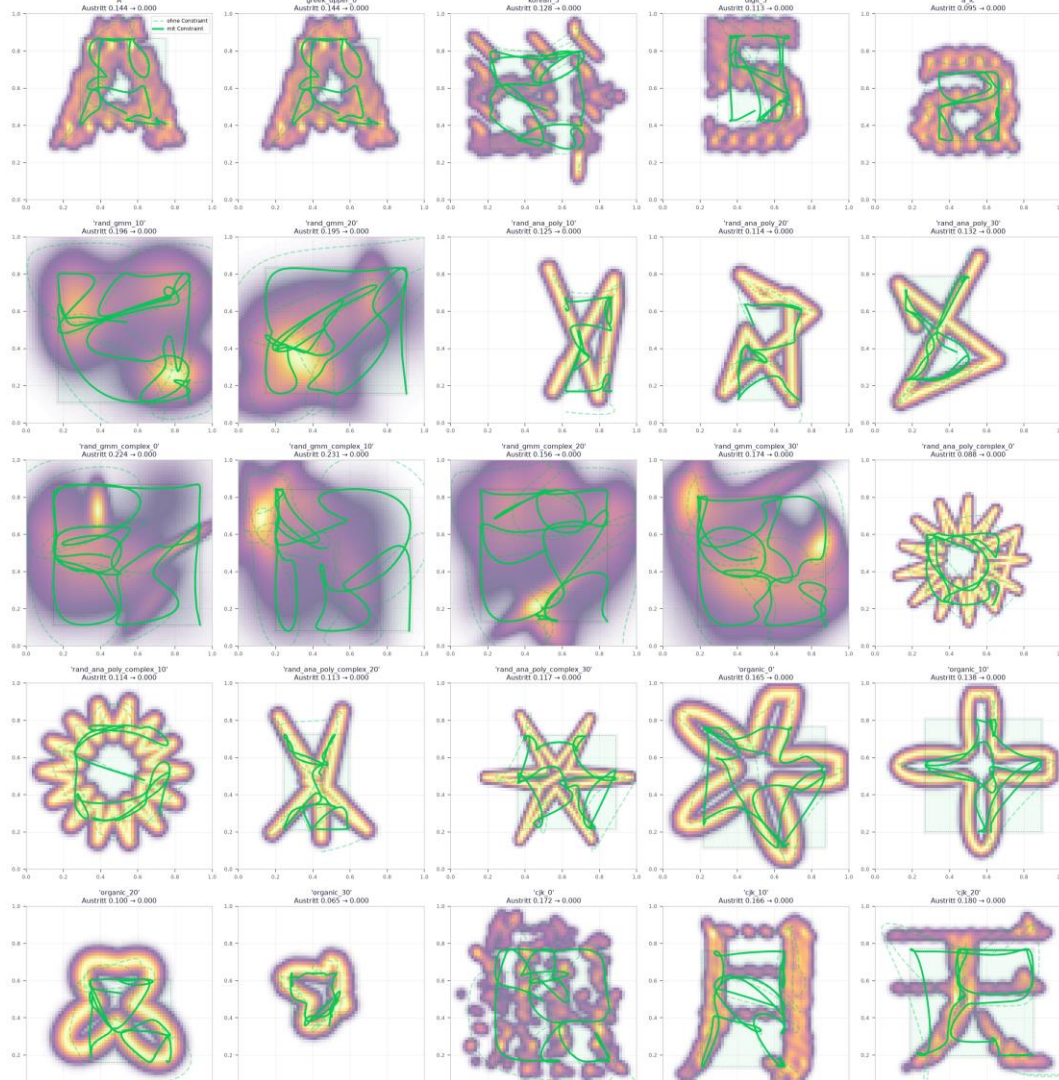
Obstacle inference

- Calculate repulsion force during each euler step during inference
- Plus further steps only for applying the obstacle force

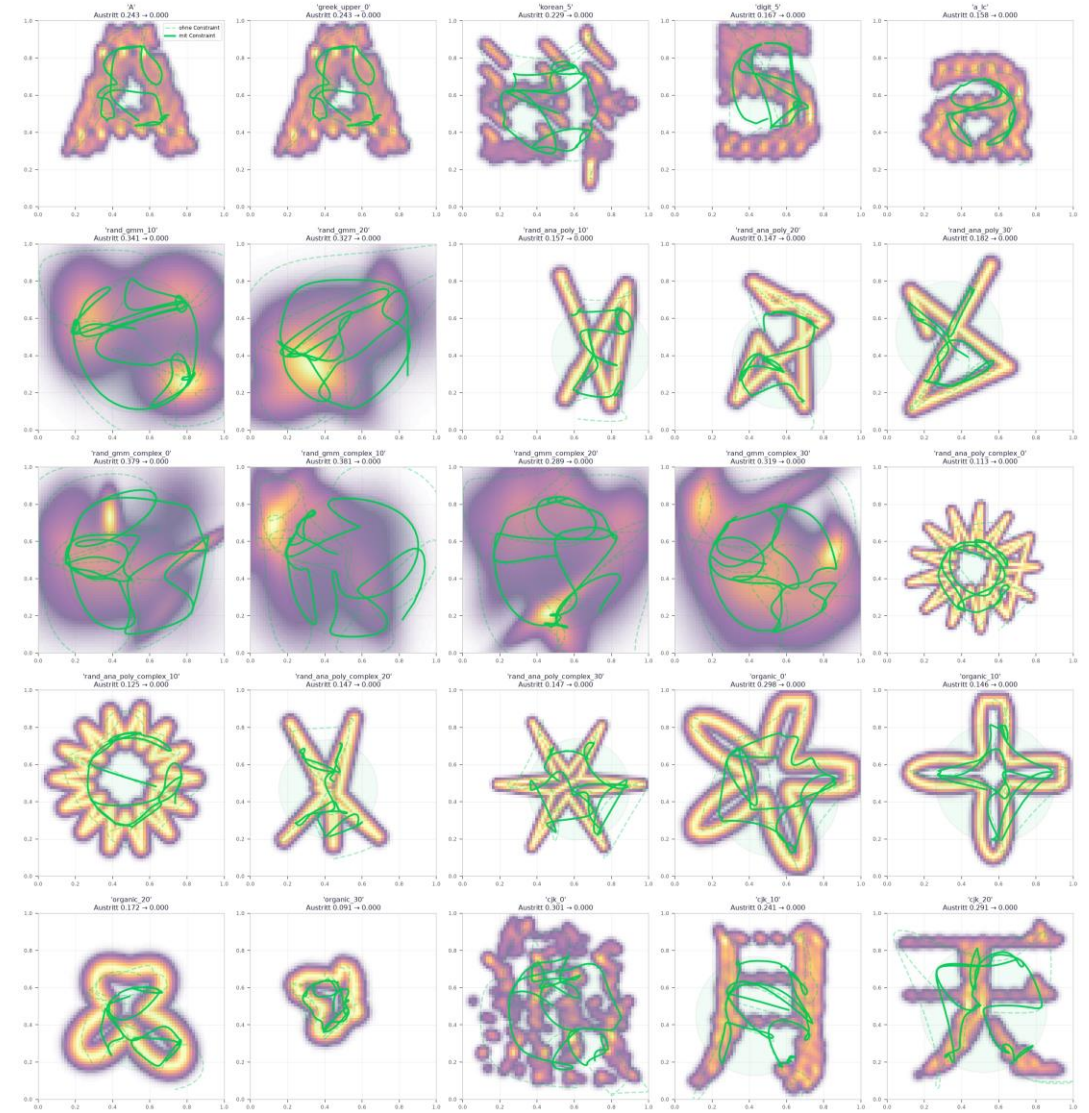


Constraints: Bounding box constraints

Constraint 1 — Keep-in-Region (box, 0.7x freie Ausdehnung)
Holdout n=25 E_erg 2.54 → 3.55 coverage 0.0411 → 0.0536

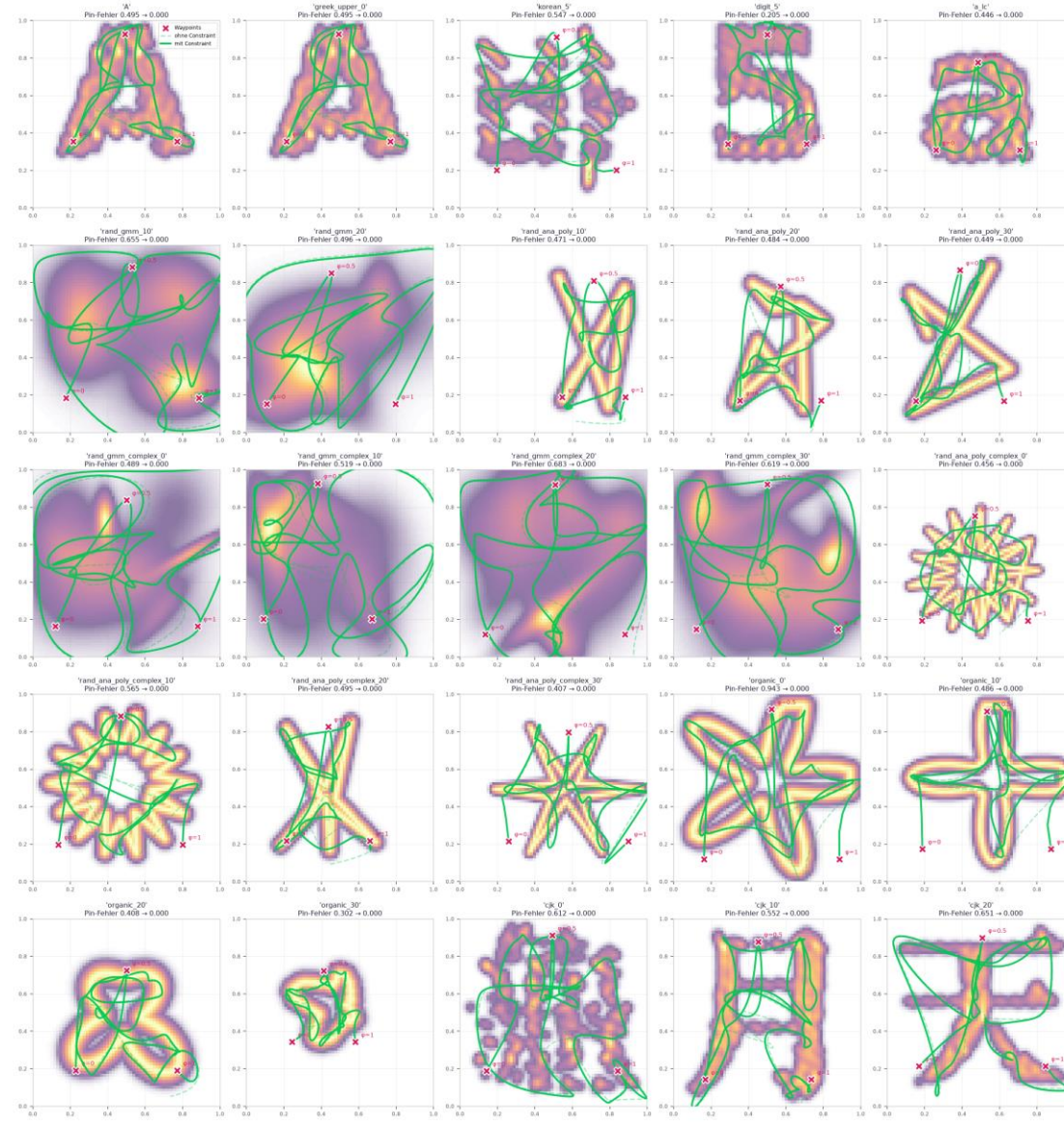


Constraint 1 — Keep-in-Region (circle, 0.7x freie Ausdehnung)
Holdout n=25 E_erg 2.54 → 5.22 coverage 0.0411 → 0.0643



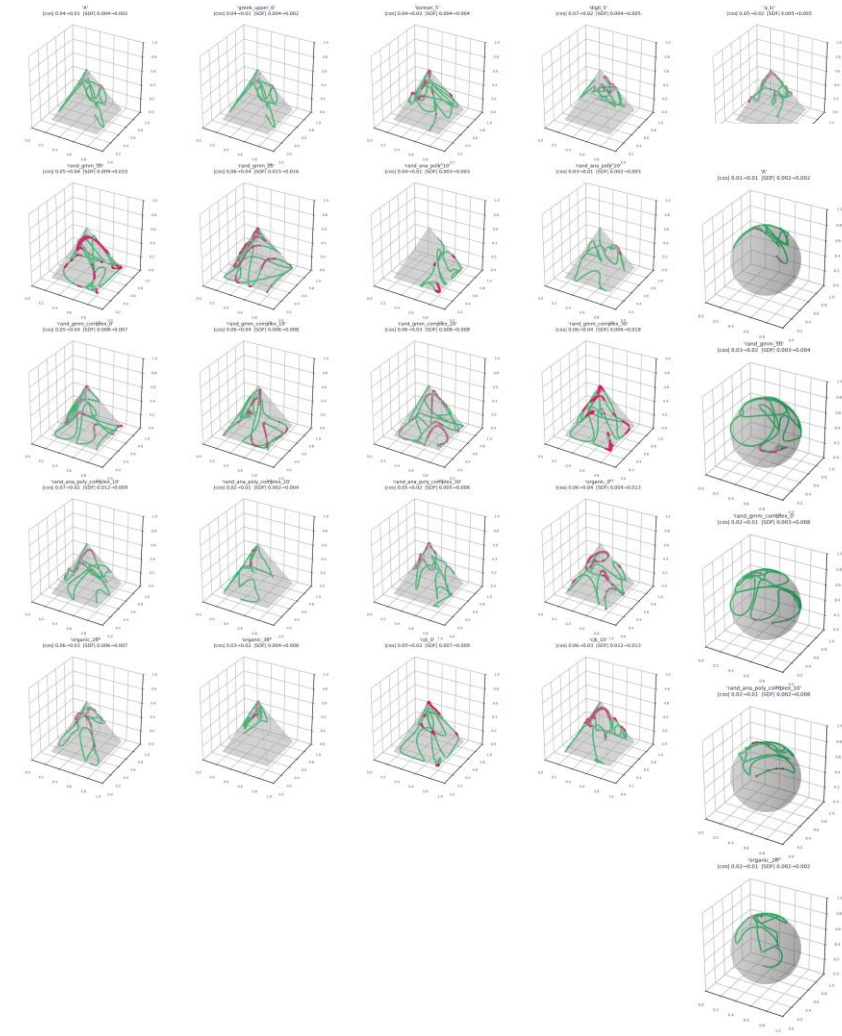
Constraints: Fix point constraint

Constraint 2 — Waypoint-Pinning (start + via + end)
Holdout n=25 E_{erg} 2.54 $\rightarrow$ 3.00 coverage 0.0411 $\rightarrow$ 0.0406



3D object penetration constraints

Constraint 5 — SE(3)-Tangentenausrichtung auf Pyramide (gestrichelt: nur Position, durchgezogen: + Ausrichtung)
[cos(t,n)] 0.048 → 0.023

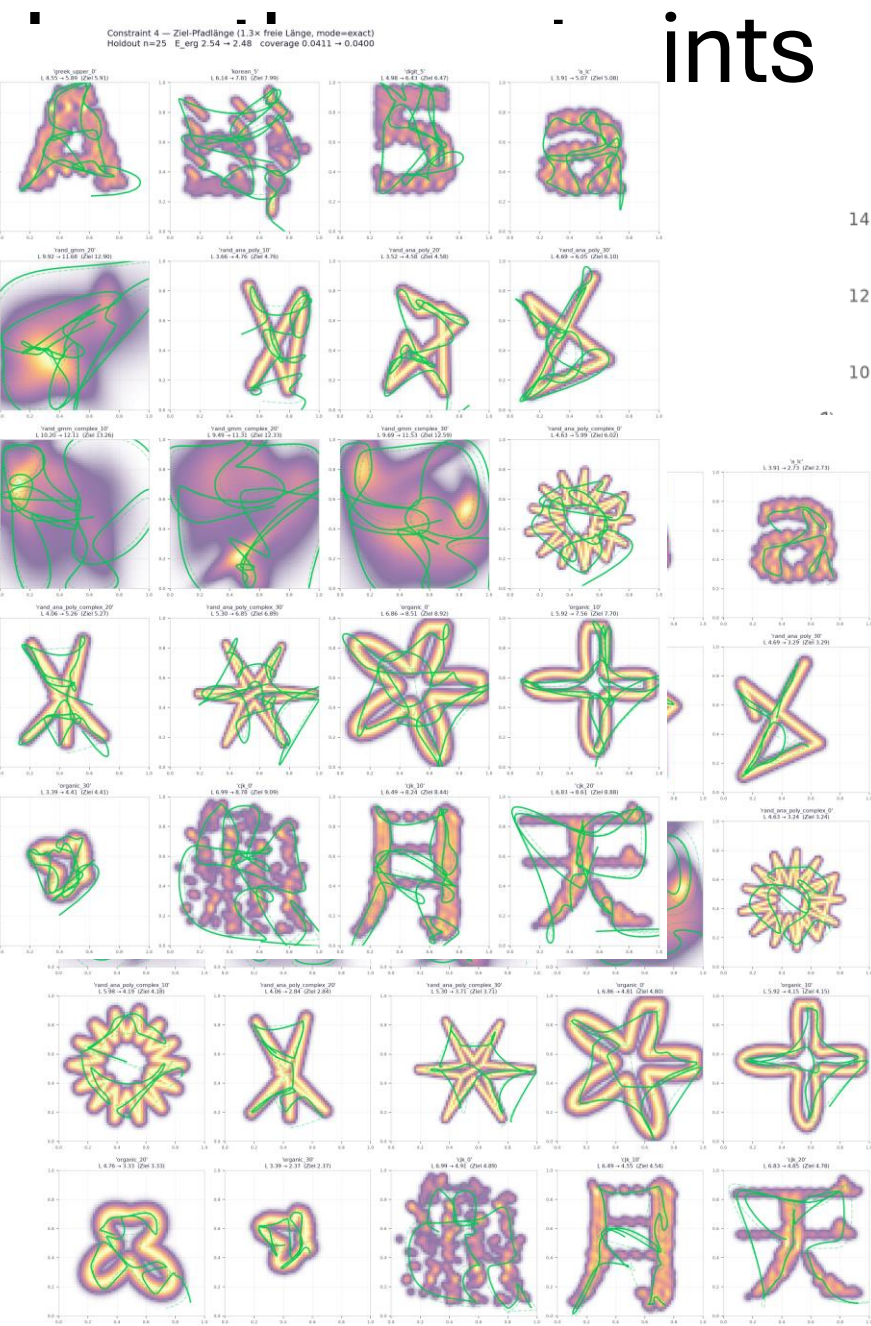


Constraint 5 — SE(3)-Tangentenausrichtung auf Sphere (gestrichelt: nur Position, durchgezogen: + Ausrichtung)
[cos(t,n)] 0.019 → 0.012

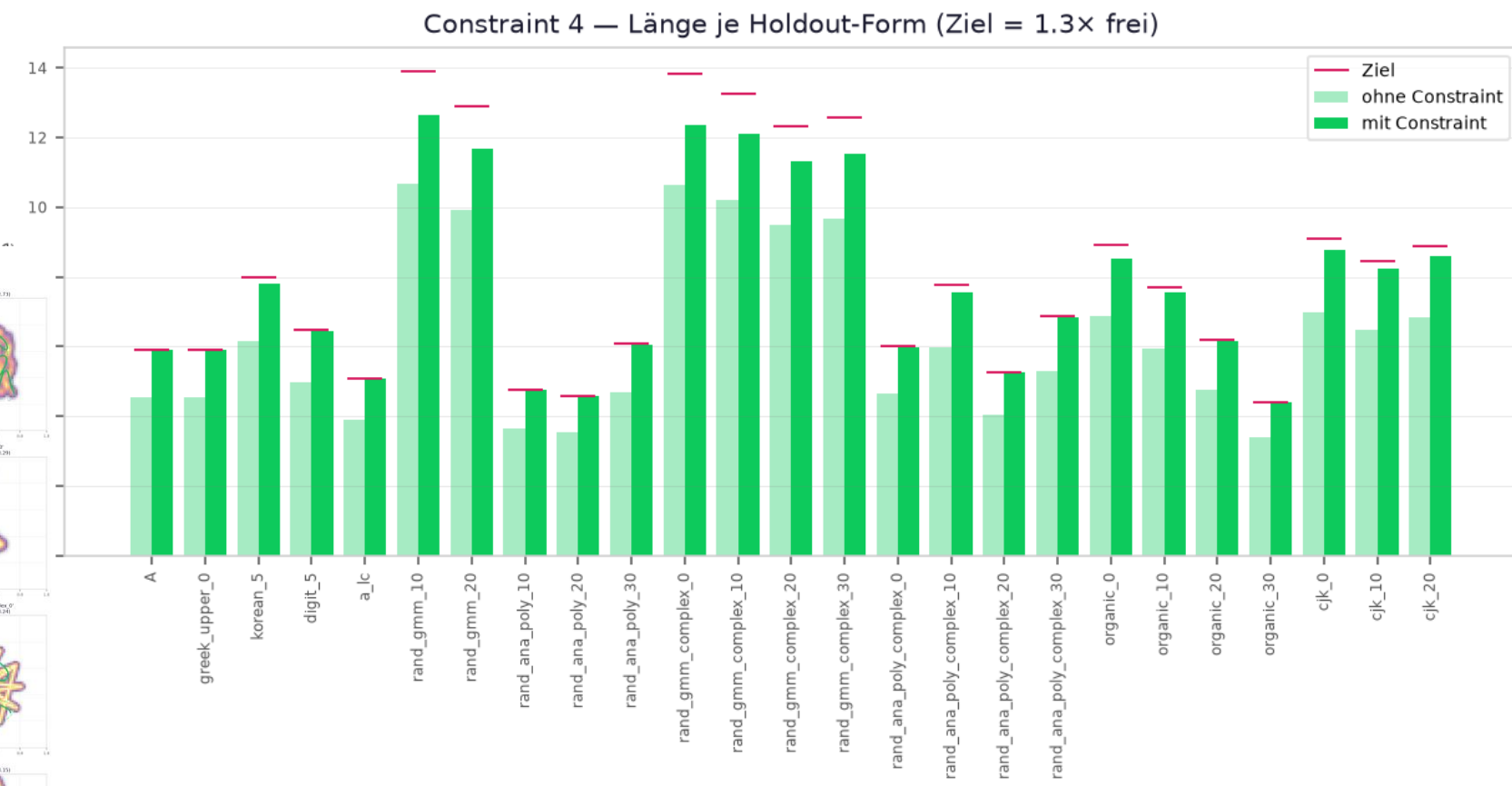


Constraint 5 — SE(3)-Tangentenausrichtung auf Torus (gestrichelt: nur Position, durchgezogen: + Ausrichtung)
[cos(t,n)] 0.023 → 0.013





ints



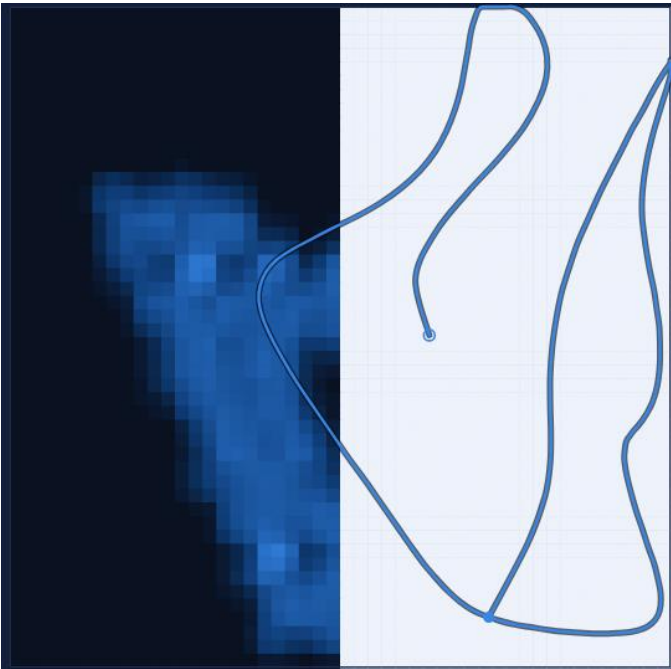
Exploration Exploitation

Belief as Target Density

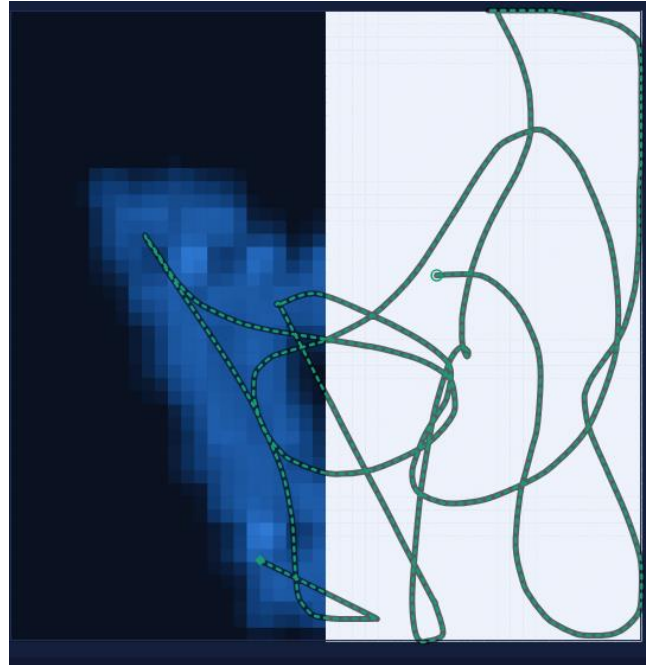
- Combine two target distributions each represented as GP:
 - $\mu(x)$:= known target distribution
 - $\sigma(x)$:= uncertainty distribution
 - $\Phi(x) = \mu(x) + \kappa \cdot \sigma(x)$
- Different planning approaches:
 - Ergodic planning on $\Phi(x)$
 - First step exploration of $\sigma(x)$, update $\mu(x)$ then exploitation of $\mu(x)$
 - Or compromiss:
 - Ergodic planning on $\Phi(x)$
 - Execute part of path, update $\Phi(x)$ replan and continue
 - ➔ Problem: no startpointconditioning
 - Trained on new databank with random startpoints and included startpointconditioning via FILM architecture (same as time conditioning)

Ergodic planning on UCB

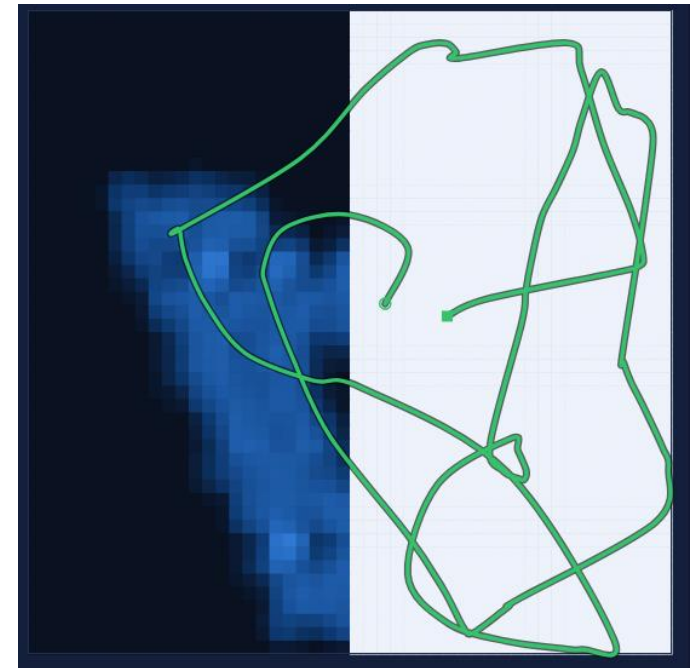
One time planning



Two stage planning



Constant replanning

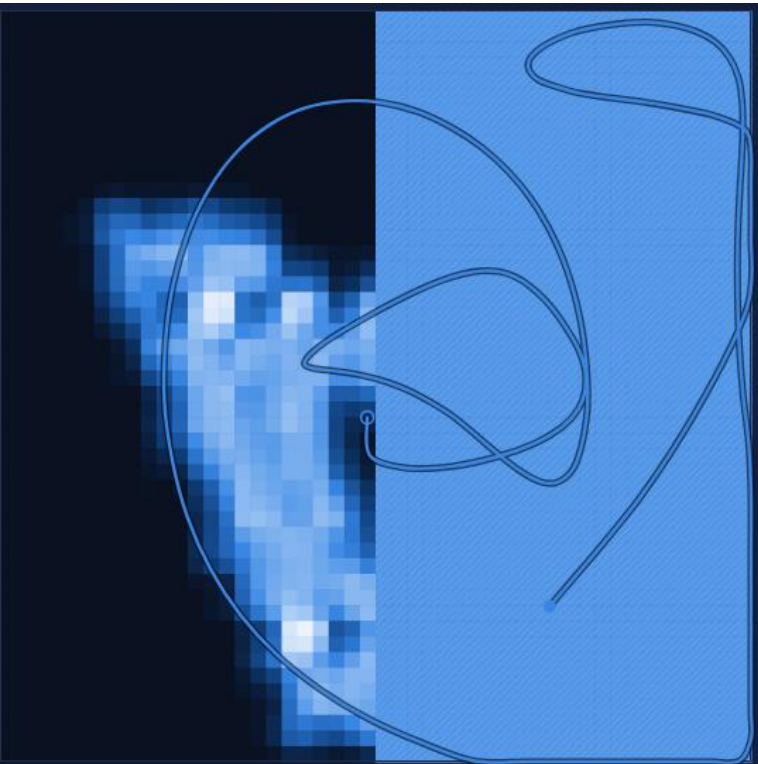


Belief as Target Density

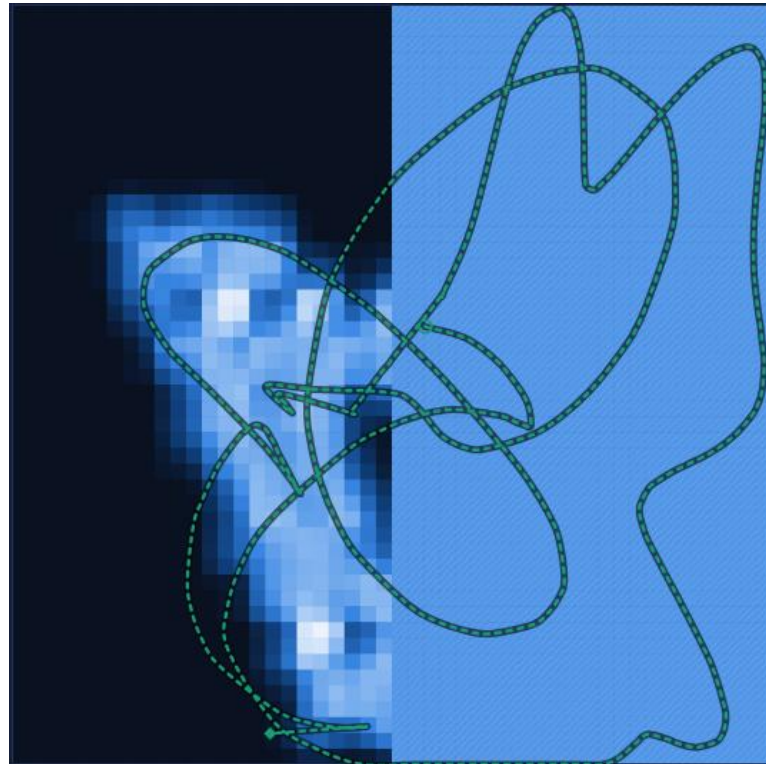
- Test different target density modelation
 - Mass equalization: $\Phi(x) = (1-w) \mu(x) + w \cdot \sigma(x)$
 - W as time for exploration weight vs exploitation
(w=0.5 means equal exploration and exploitation)

Ergodic planning on UCB with Mass equalization

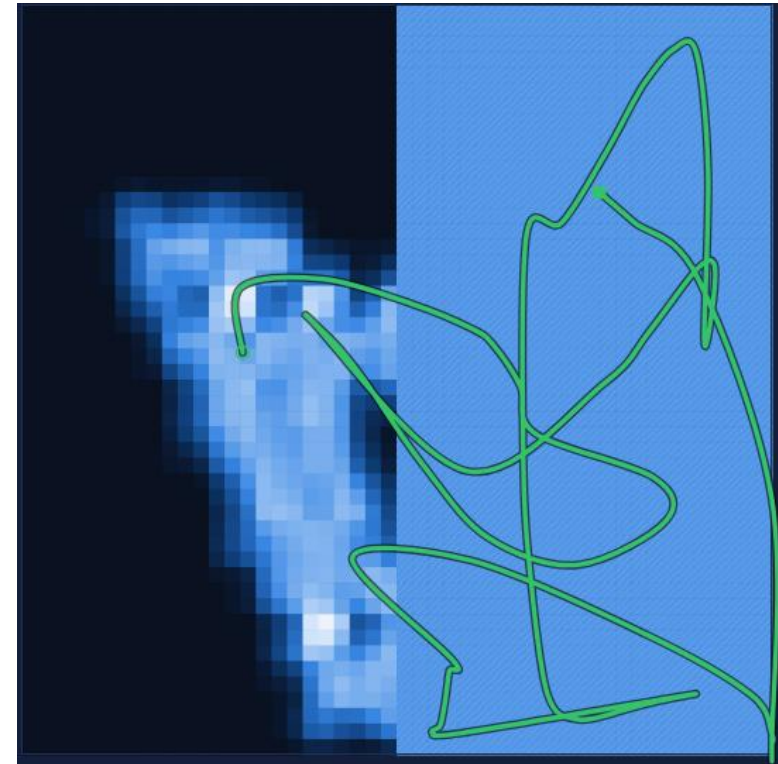
One time planning



Two stage planning



Constant replanning

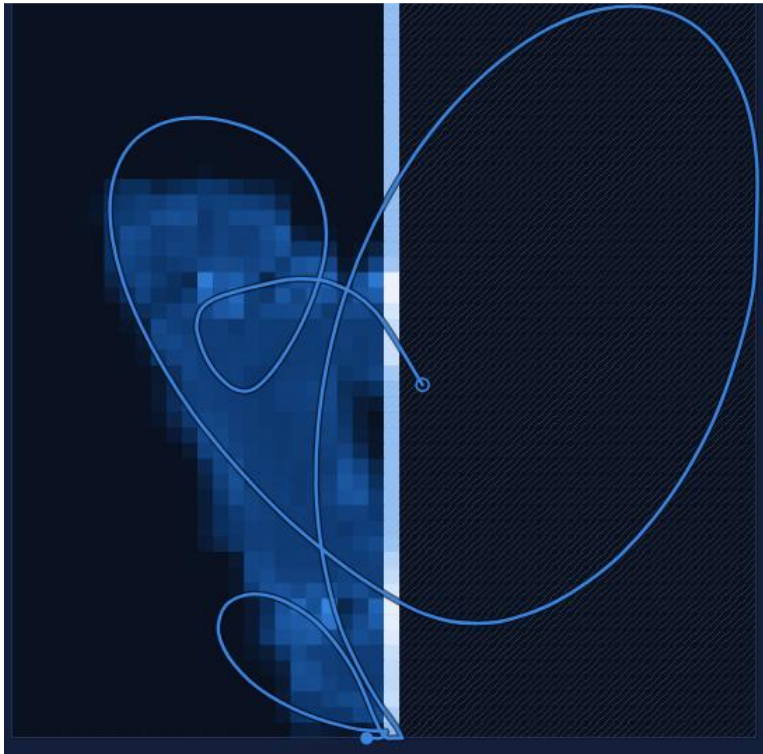


Belief as Target Density

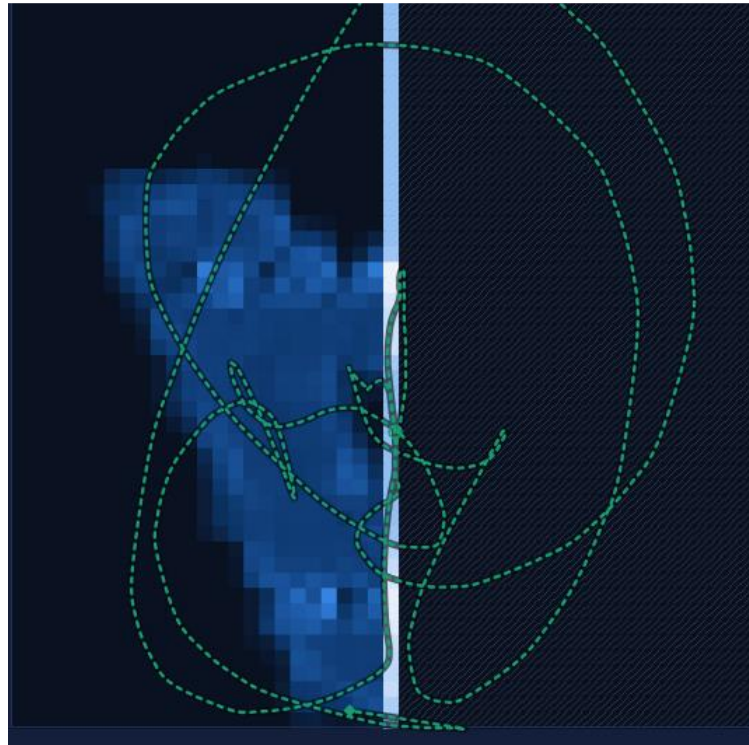
- Fisher-Information idea from paper “Ergodic Exploration of Distributed Information”
 - Idea: Where will another measurement deliver the most information gain?
 - Information gain or change in information density: $\|\nabla\mu(x)\|^2$
plus where is high information density for exploration: $\hat{\mu}(x)$
 - Math: $\Phi(x) = \hat{\mu}(x) + \kappa\|\nabla\mu(x)\|^2$

Ergodic planning on Fisher-Information

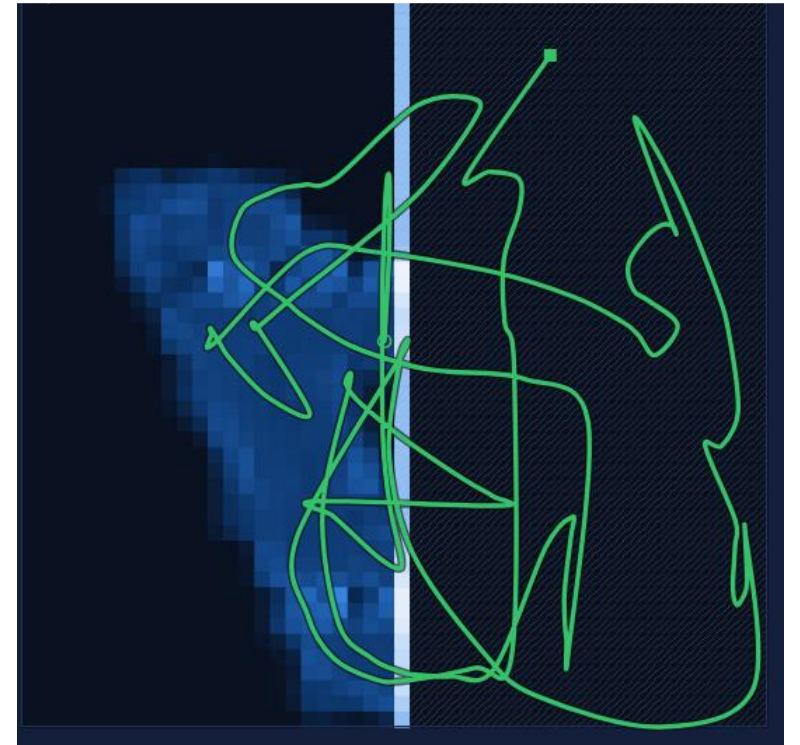
One time planning



Two stage planning



Constant replanning

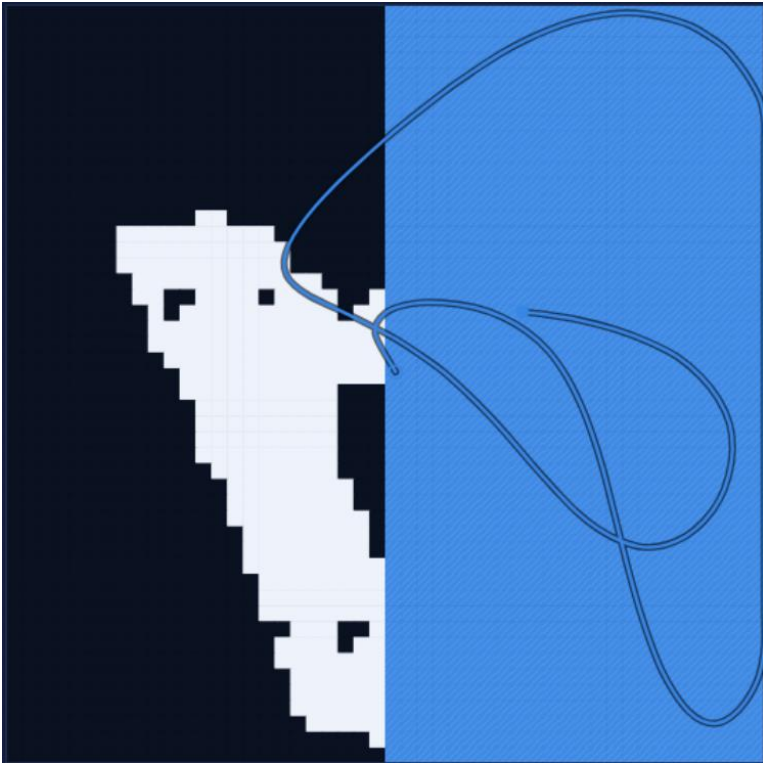


Belief as Target Density

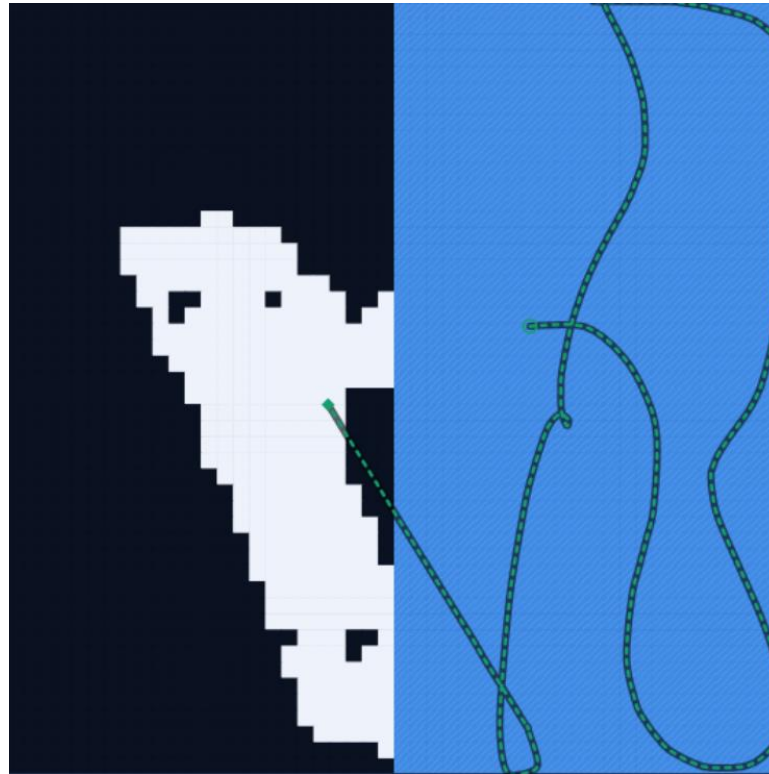
- **LSE (Level Set Estimation)**
 - Probability of being part of the target distribution
 - $\text{Phi_LSE}(x) = P(f(x) > \tau)$
 - $z(x) = (\mu(x) - \tau) / \max(\sigma(x), 1e-9)$
 - Erf := gaussian error funktion
 - $\text{Phi_LSE}(x) = 0.5 * [1 + \text{erf}(z(x) / \sqrt{2})]$

LSE Niveau

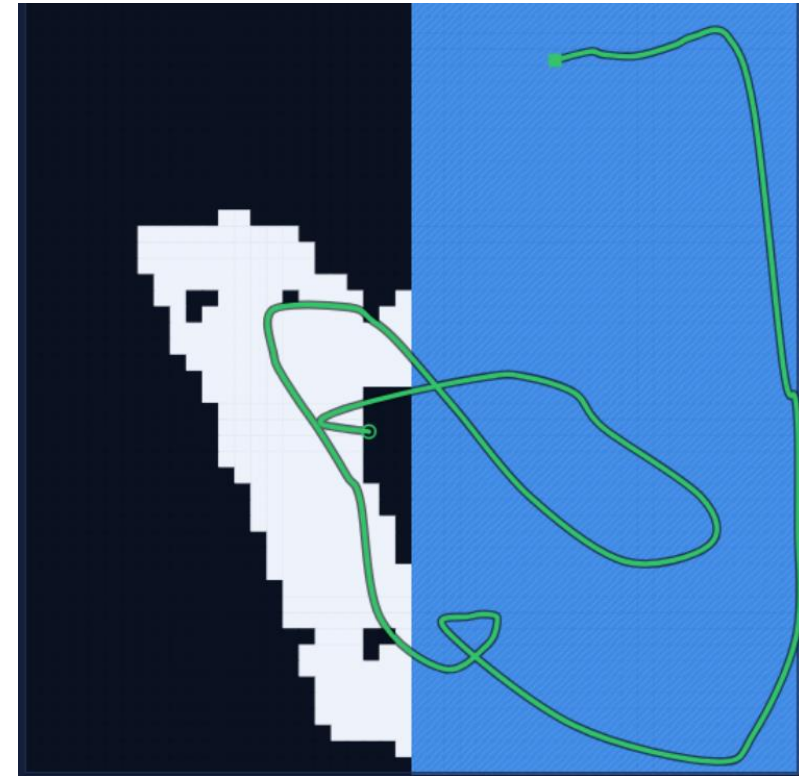
One time planning



Two stage planning



Constant replanning



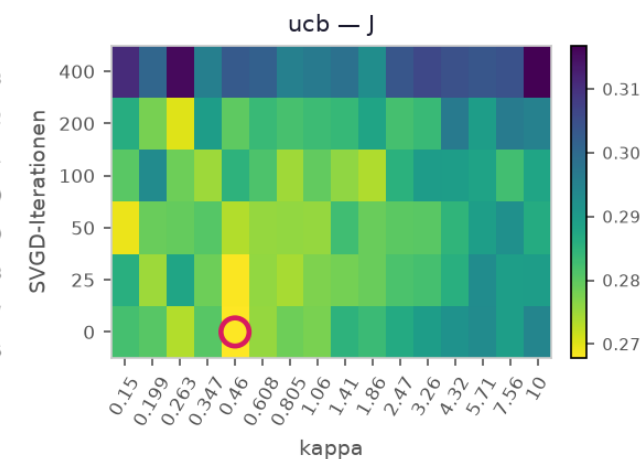
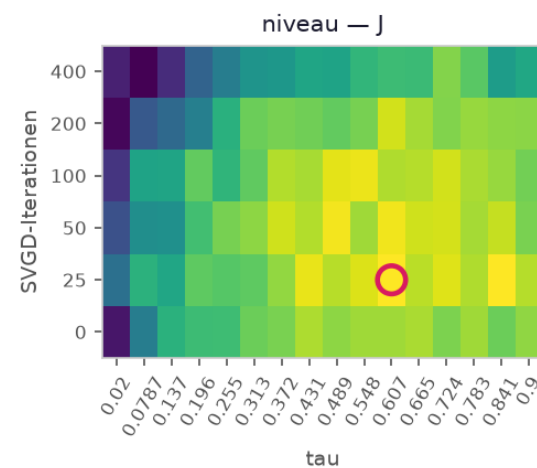
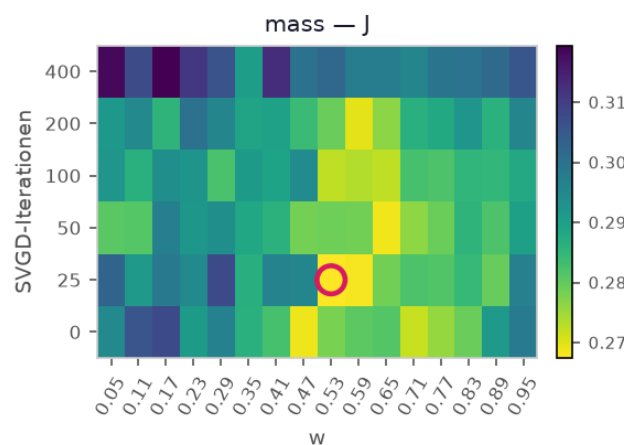
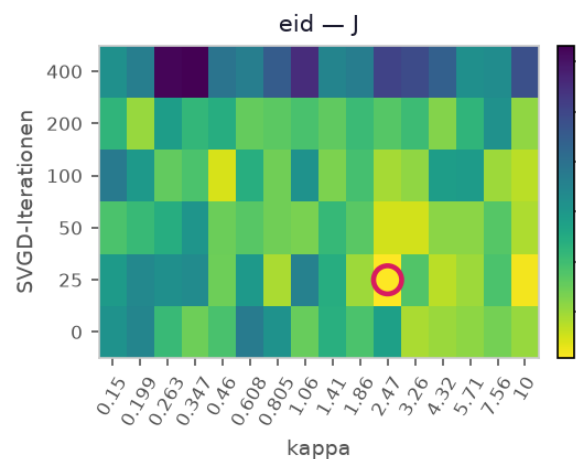
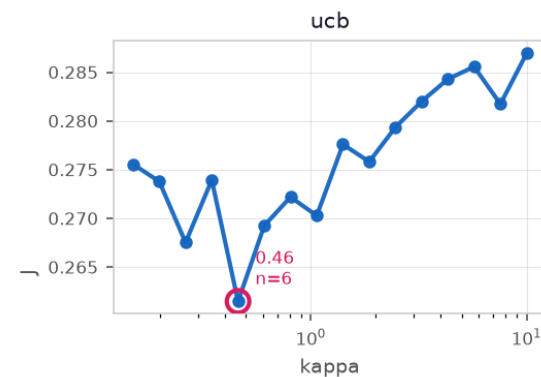
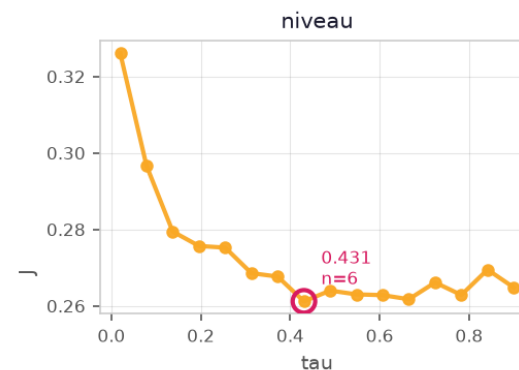
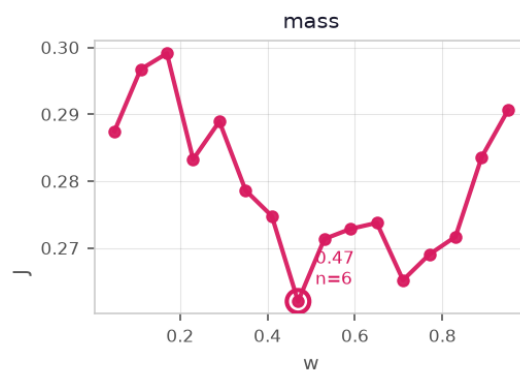
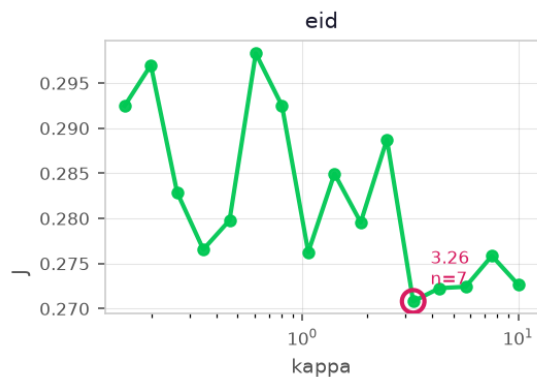
Find ideal exploration/exploitation setup

- 1. Hyperparameter optimization
- 2. simple NN to learn hyperparameters
- 3. Reinforcement learning for hyperparameters

Paramter optimization

- Evaluation function:
 - $J = q(n) + 0,02 \cdot n + 0,004 \cdot t(n)$
 - $q(n) :=$ coverage error on ground truth target
 - $n :=$ number of pathunits (Pathunit is diagonal of workspace)
- Grid based search (target distributions x paramtervalue x num_svdg): List of parameter combination and further variations at the best points.

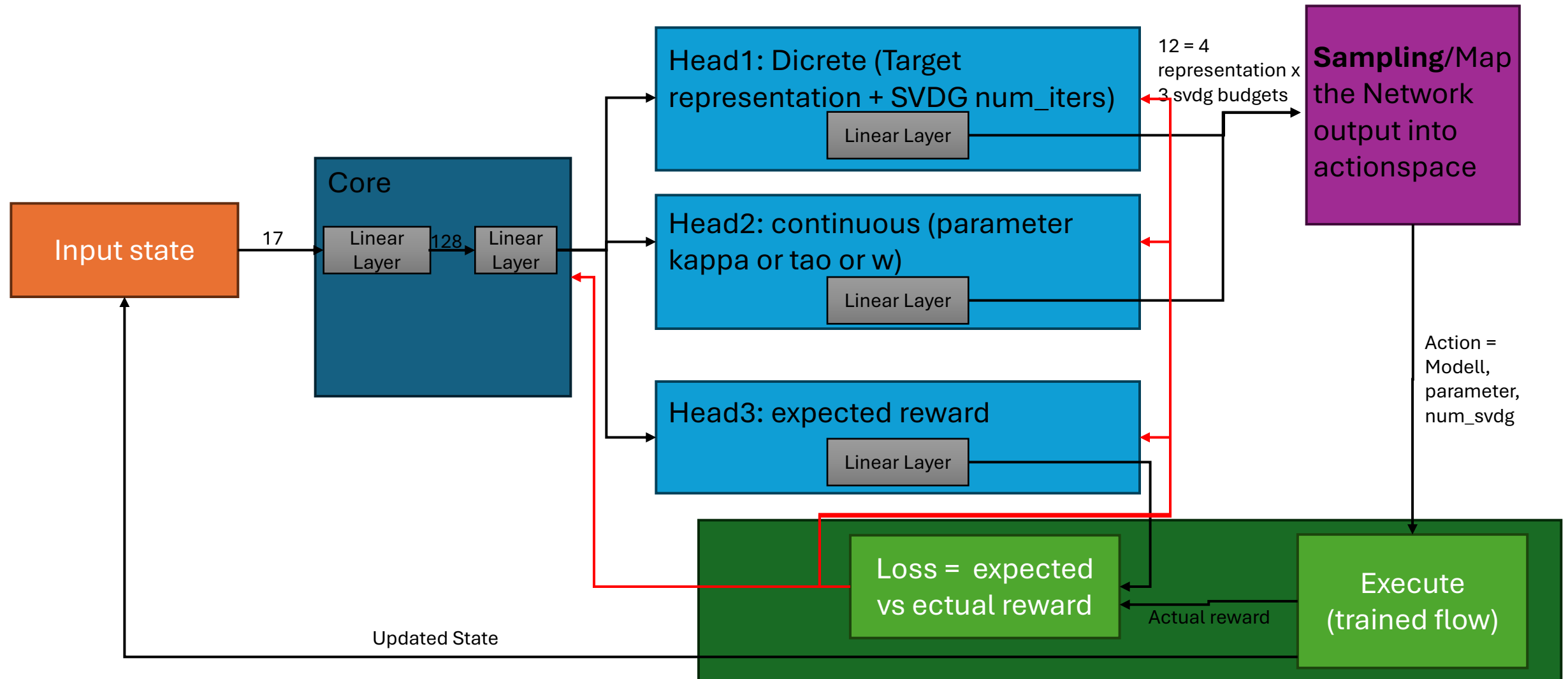
Results Paramter optimization

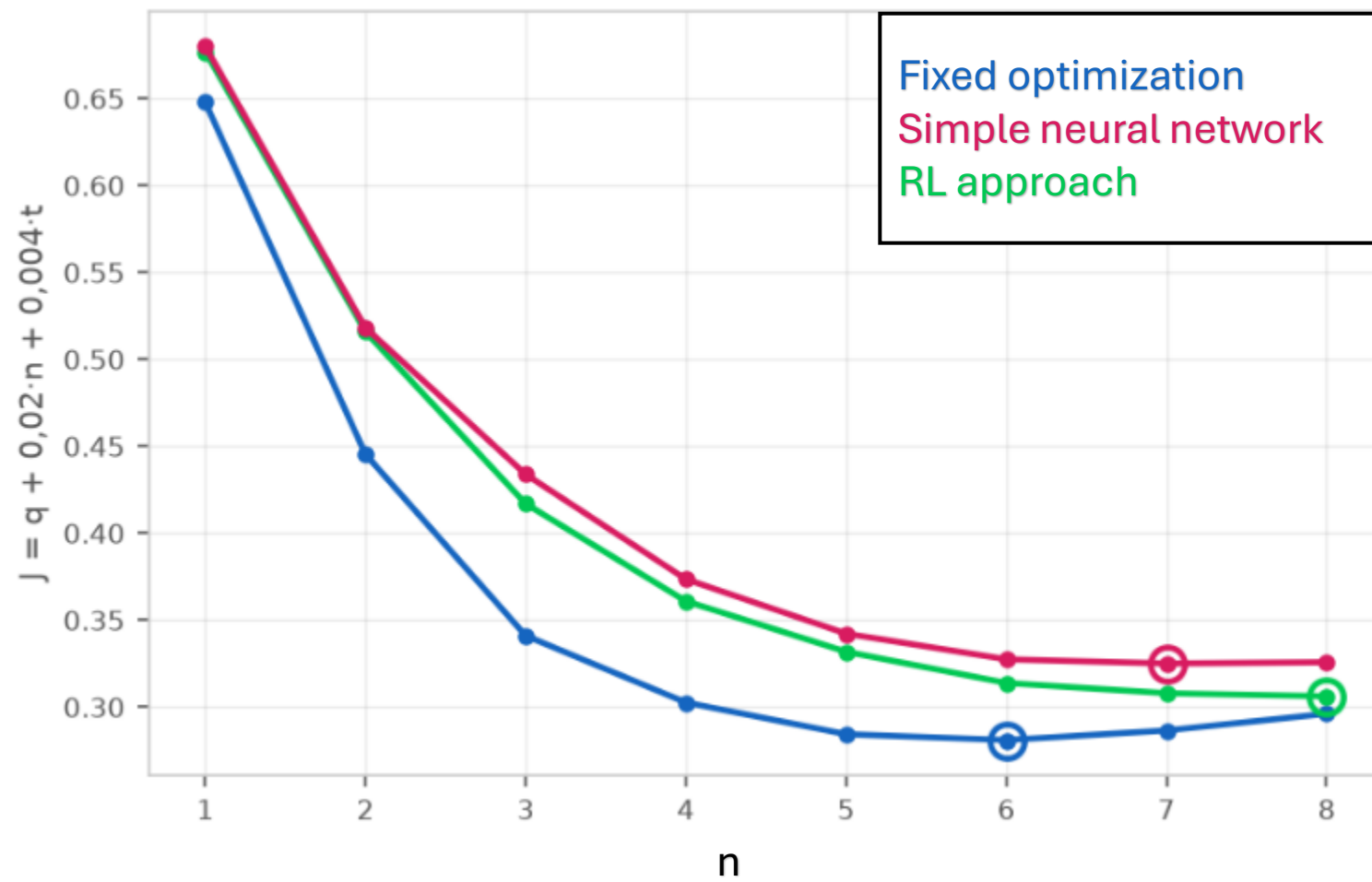


Learned Parameters

- Generate trajectories based on parametergrid, evaluate them on J
- => Datapair: parameters and J
 - Train simple nn on parameterpair: 3 linear layers

RL Approach - Architecture





GUI interactive demo

Research Questions

- **Ergodic improvement**

- How can we use flow matching to enhance ergodic coverage algorithms?
- How many svdg iterations can be saved by learning a prior?

- **Constraints**

- How many/which constraints can be enforced by learning conditions vs during inference?

- **Exploration/Exploitation**

- How can we best represent the target distribution which allows both exploitation of known targets and exploration of unknown areas?
- How can we learn an exploration/exploitation tradeoff?

- **2D/3D**

- How well can a Unet flowmatching network generalize in 2D vs 3D?

Thesis title

- Learned smooth trajectories for ergodic exploration (and exploitation)

Next week

- Possible: Try larger dataset for NN parameter learning and Reinforcement learning
- Test 3D constraints (orientation 3D obstacles etc)
- Prepare for lab tests